

Biscayne Bay Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 05 May, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

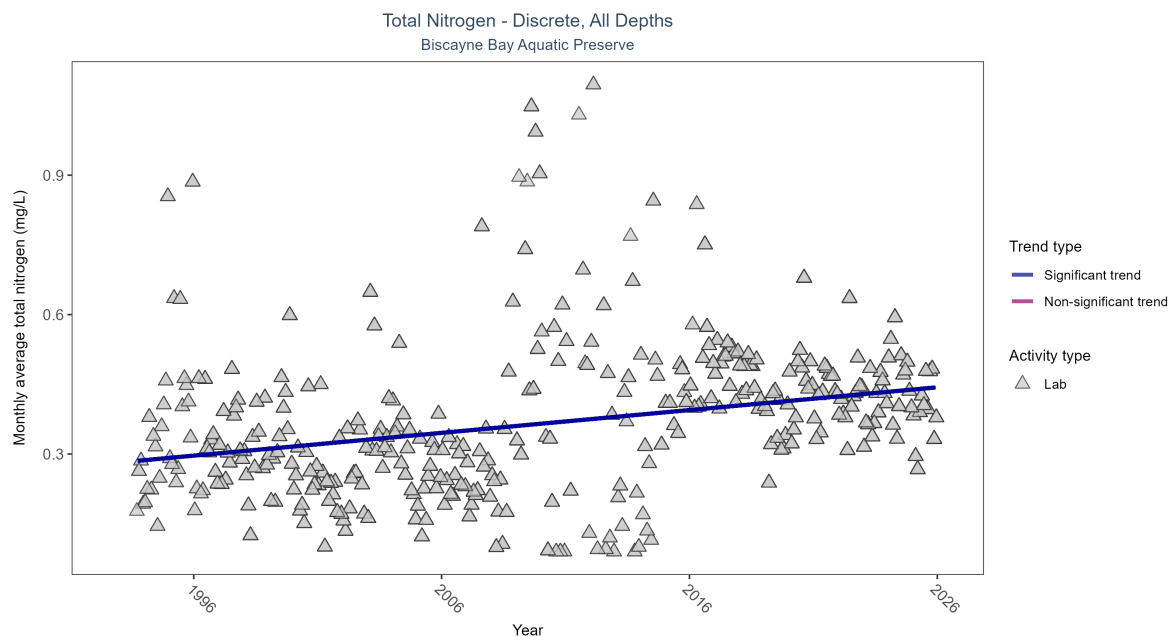


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	9363	33	1993 - 2025	0.335	0.23873	0.2818	0.0049	0

Monthly average total nitrogen increased by less than 0.01 mg/L per year.

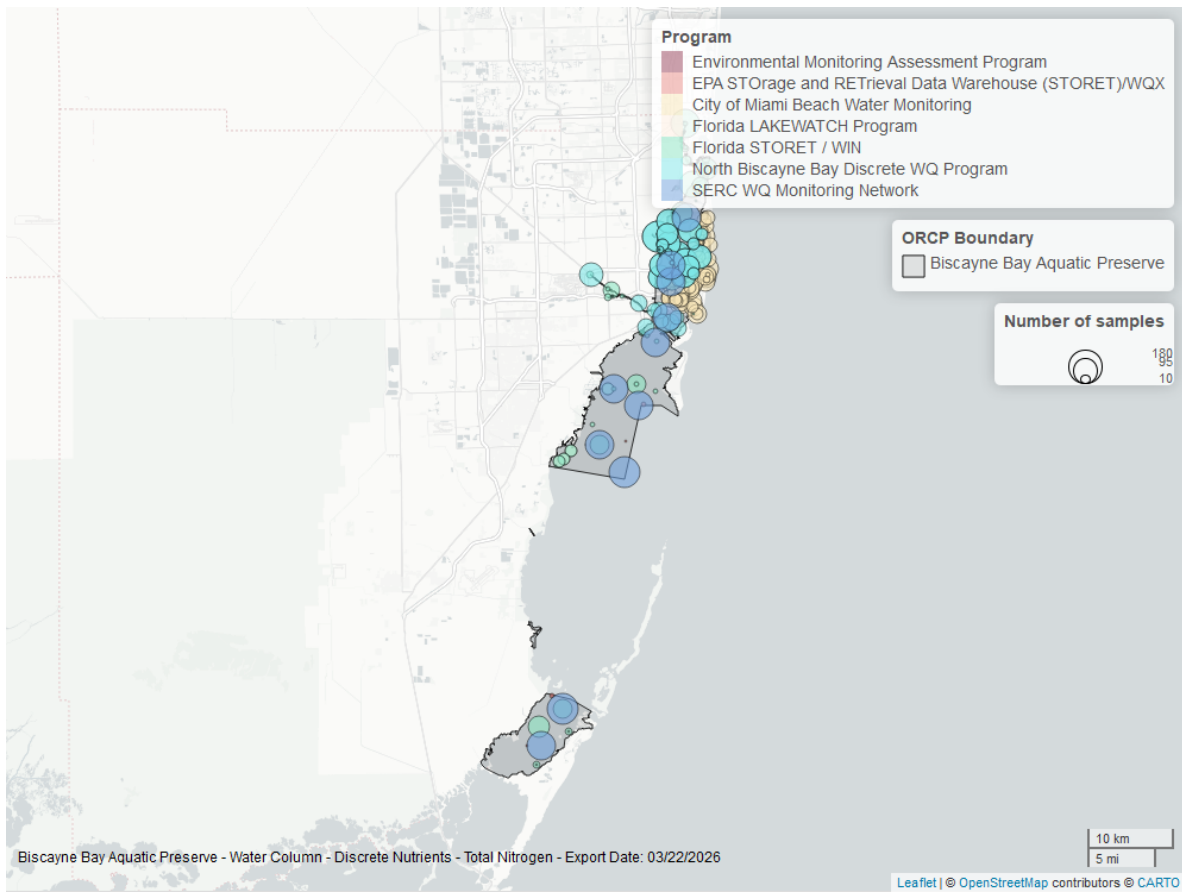


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

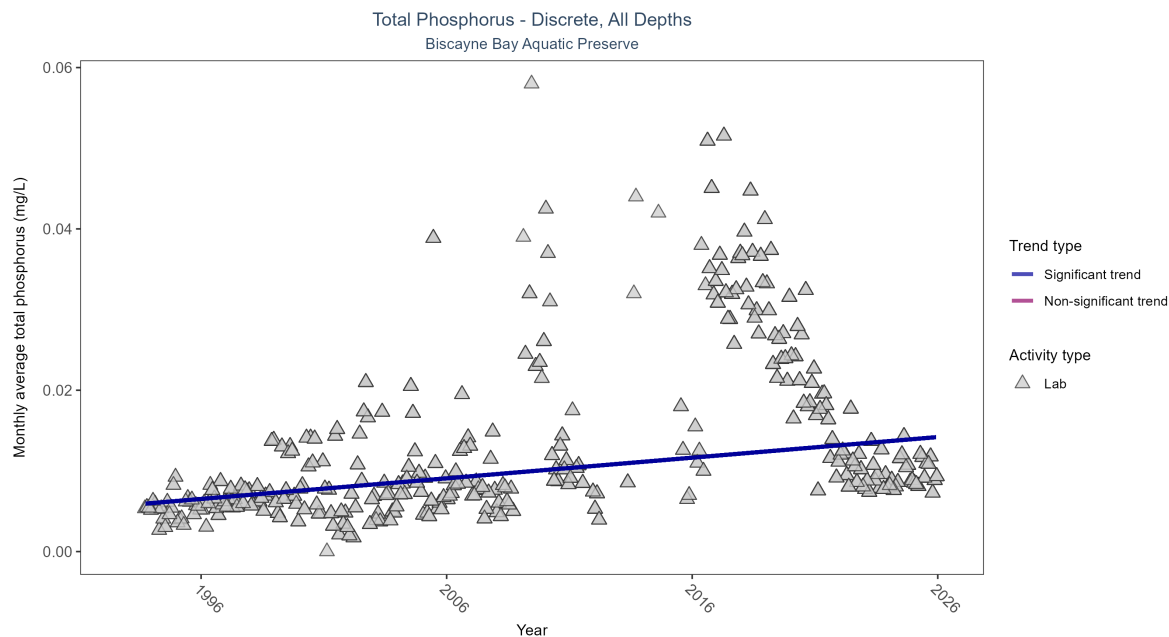


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	9858	33	1993 - 2025	0.009	0.37984	0.00576	0.00026	0

Monthly average total phosphorus increased by less than 0.01 mg/L per year.

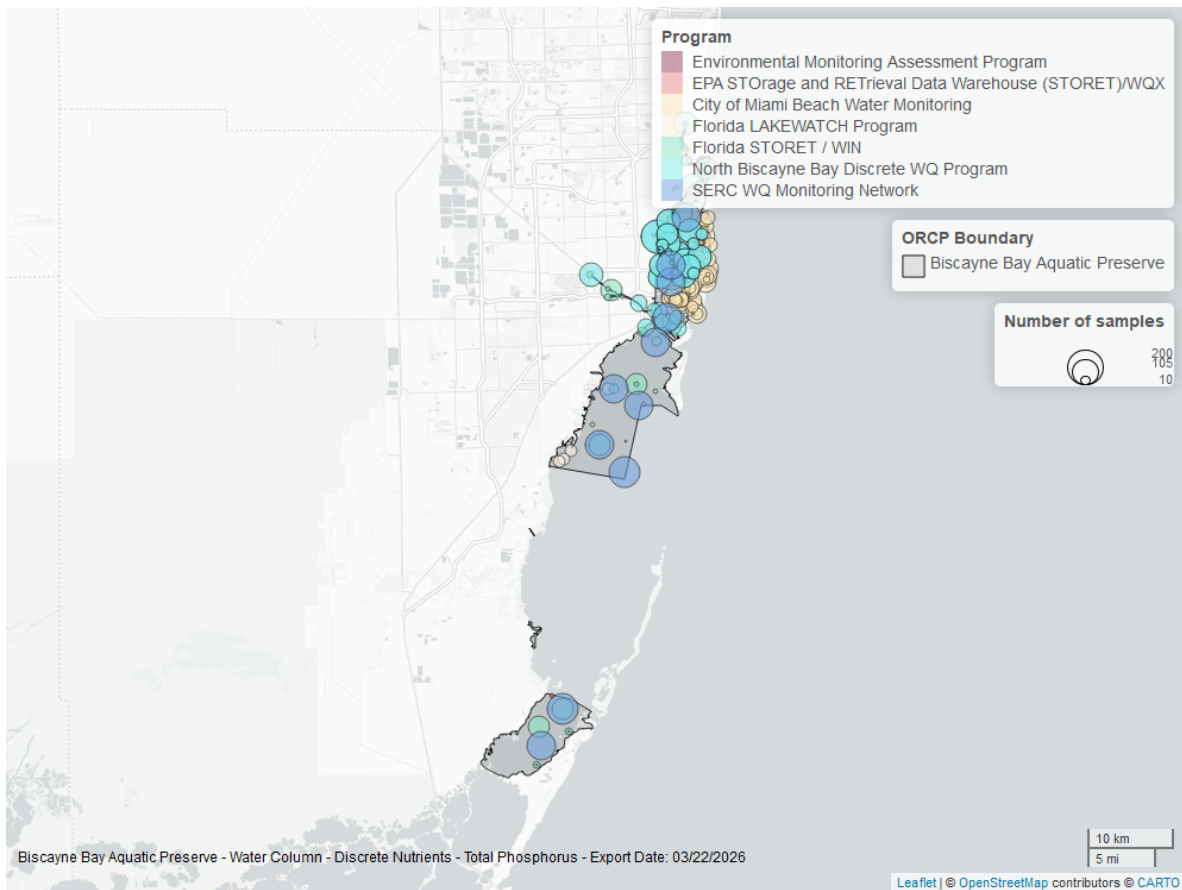


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

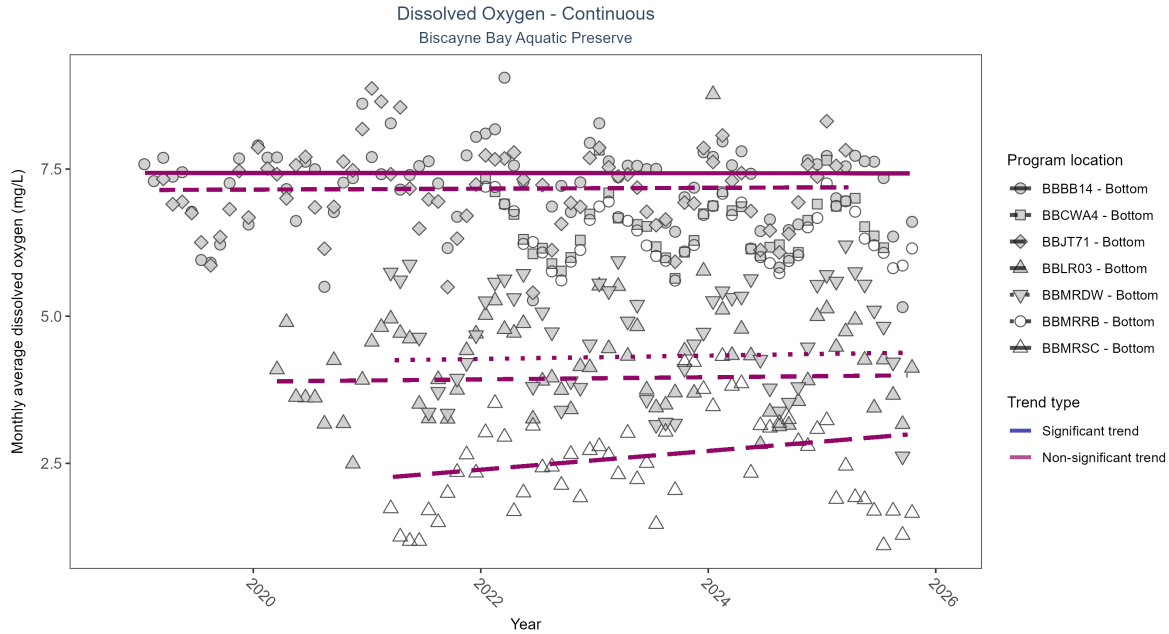


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	209766	7	2019 - 2025	7.2	-0.01	7.43	0	1
BBCWA4	Insufficient data to calculate trend	108422	4	2022 - 2025	6.5	-	-	-	-
BBJT71	No significant trend	207039	7	2019 - 2025	7.0	0.06	7.14	0.01	0.5599
BBLR03	No significant trend	177333	6	2020 - 2025	4.2	0.06	3.89	0.02	0.6828
BBMRDW	No significant trend	150197	5	2021 - 2025	4.7	0.1	4.25	0.03	0.58
BBMRRB	Insufficient data to calculate trend	124787	4	2022 - 2025	6.4	-	-	-	-
BBMRSC	No significant trend	151445	5	2021 - 2025	2.2	0.23	2.23	0.16	0.076

No detectable change in monthly average dissolved oxygen was observed at five locations. There was insufficient data to fit a model for two locations.

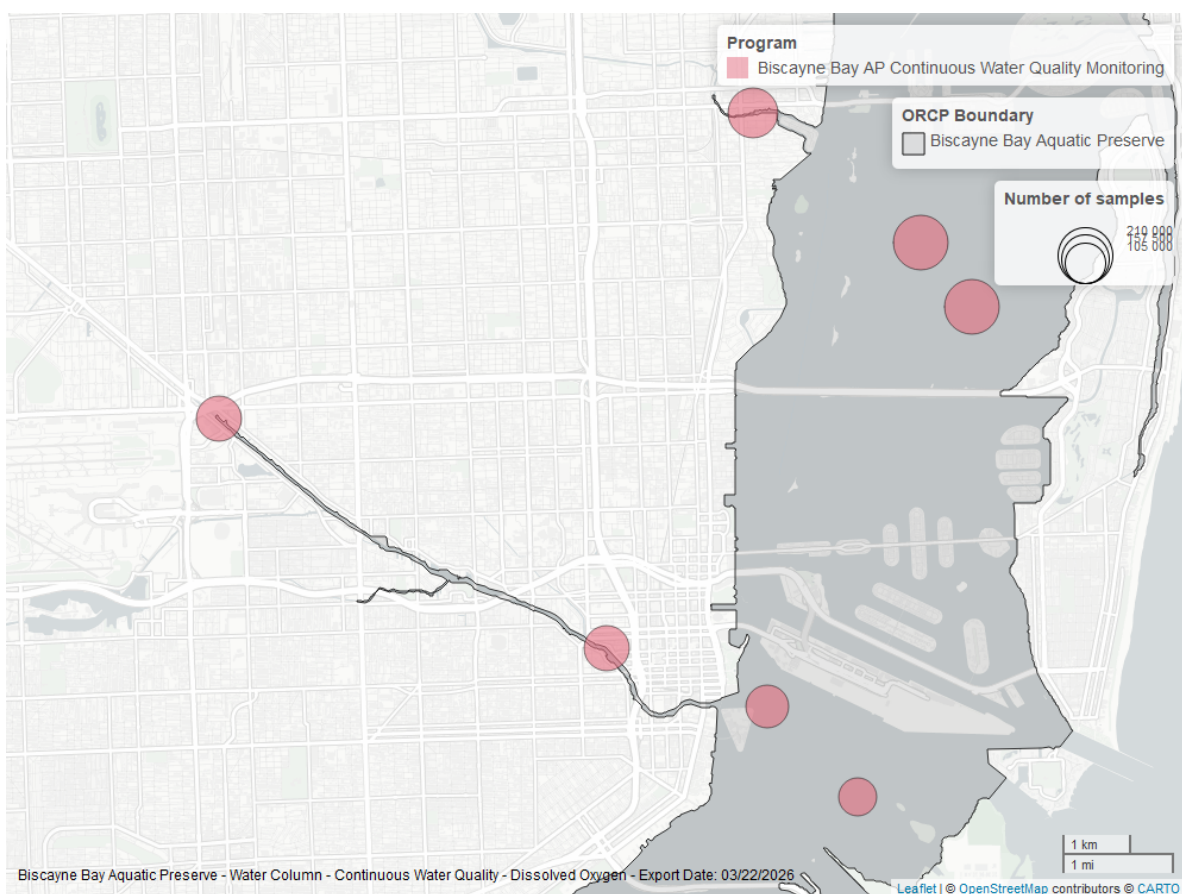


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

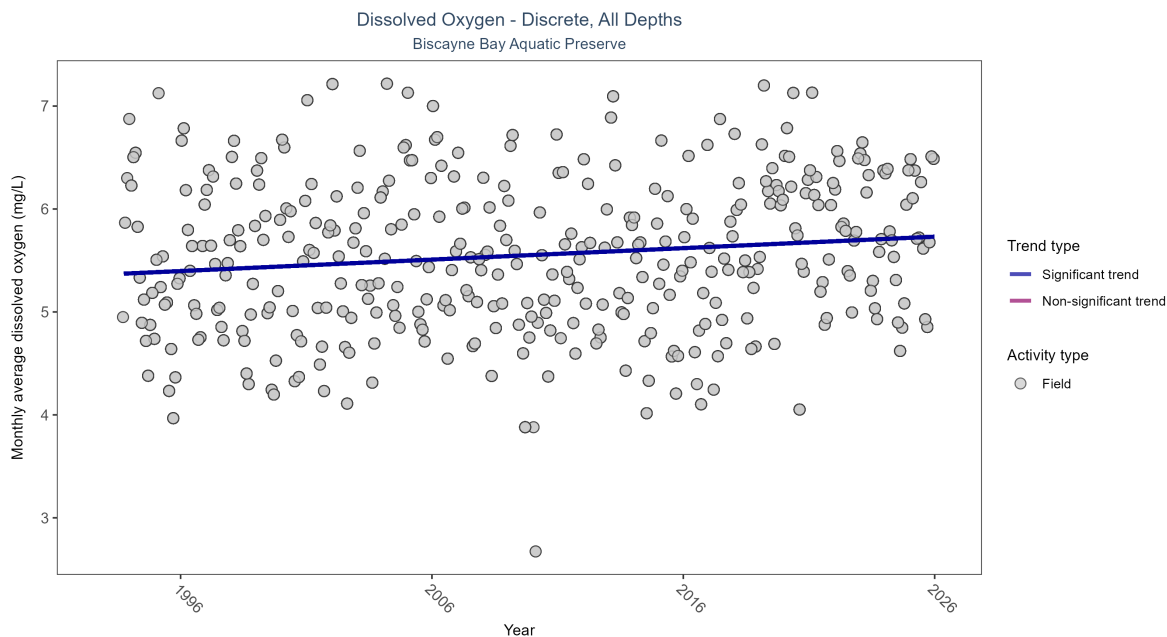


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	45505	33	1993 - 2025	5.81	0.179	5.36365	0.01113	0

Monthly average dissolved oxygen increased by 0.01 mg/L per year.

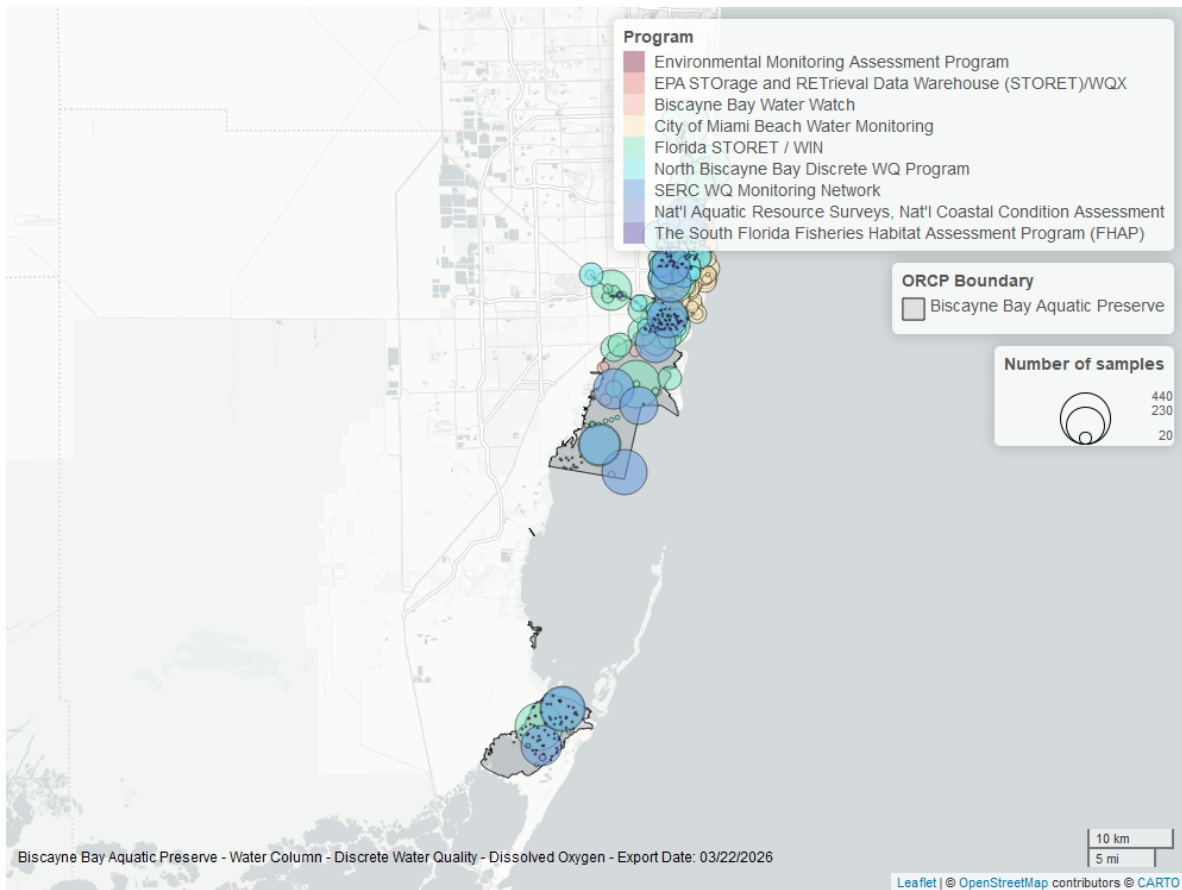


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

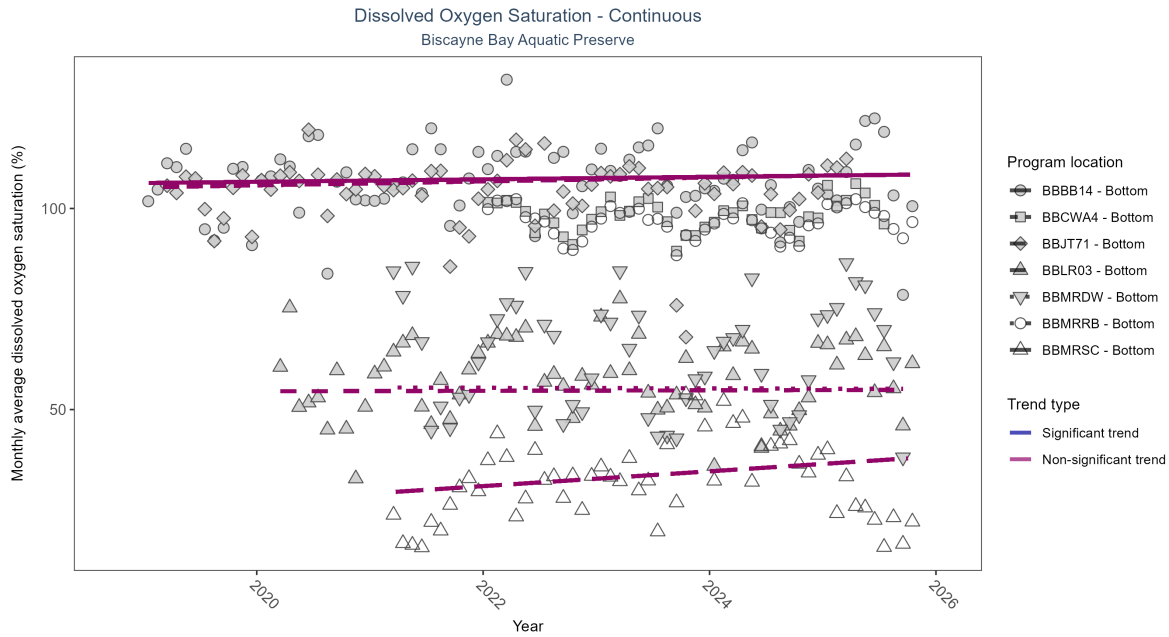


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	209506	7	2019 - 2025	104.6	0.08	106.33	0.31	0.4397
BBCWA4	Insufficient data to calculate trend	112145	4	2022 - 2025	97.8	-	-	-	-
BBJT71	No significant trend	208190	7	2019 - 2025	102.0	0.13	105.26	0.49	0.1852
BBLR03	No significant trend	177332	6	2020 - 2025	57.7	0.03	54.54	0.06	0.861
BBMRDW	No significant trend	150198	5	2021 - 2025	62.5	-0.02	55.49	-0.07	0.937
BBMRRB	Insufficient data to calculate trend	124795	4	2022 - 2025	96.1	-	-	-	-
BBMRSC	No significant trend	151340	5	2021 - 2025	29.3	0.21	29.12	1.84	0.1052

No detectable change in monthly average dissolved oxygen saturation was observed at five locations. There was insufficient data to fit a model for two locations.

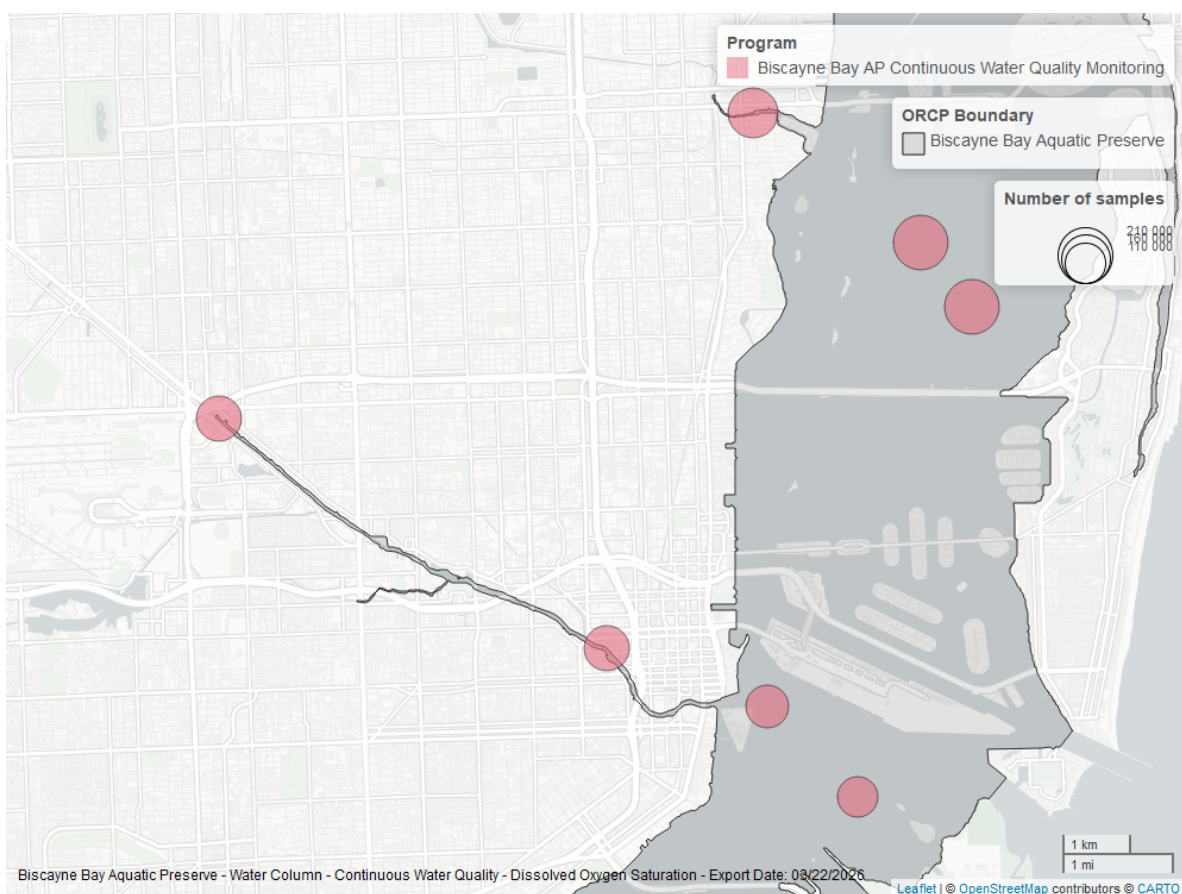


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

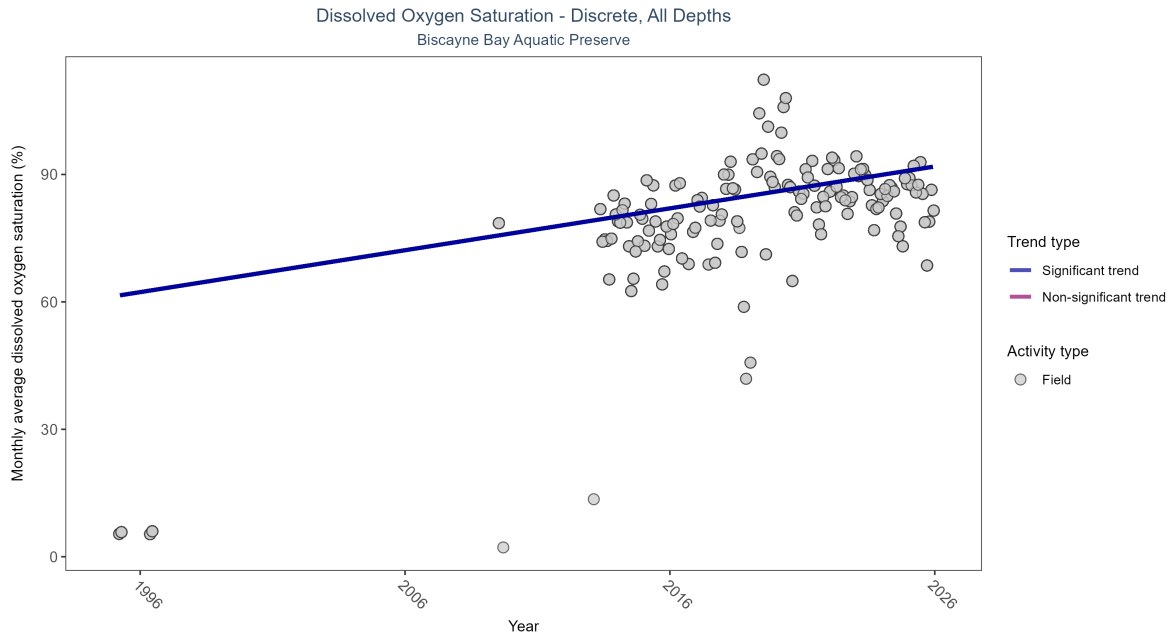


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	15093	16	1995 - 2025	88.7	0.38467	61.31281	0.9861	0

Monthly average dissolved oxygen saturation increased by 0.99% per year.

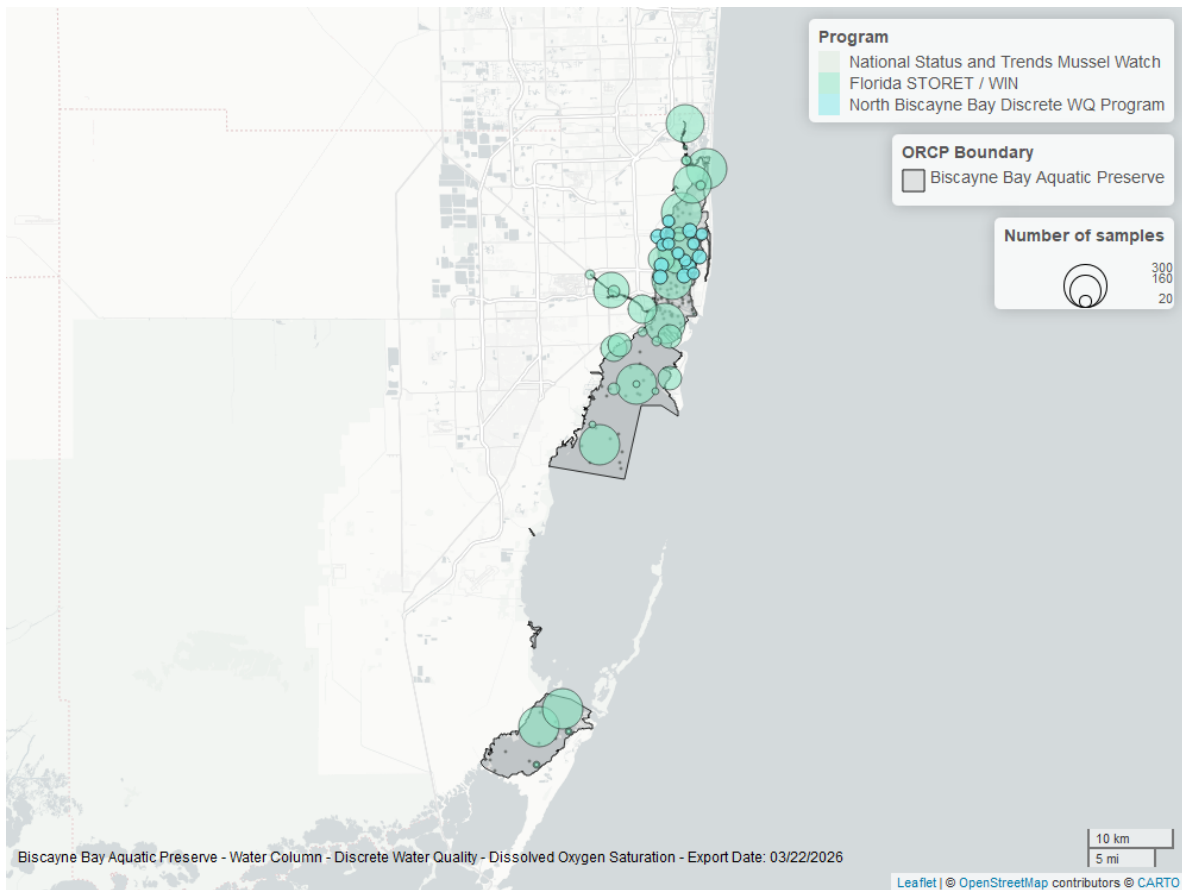


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

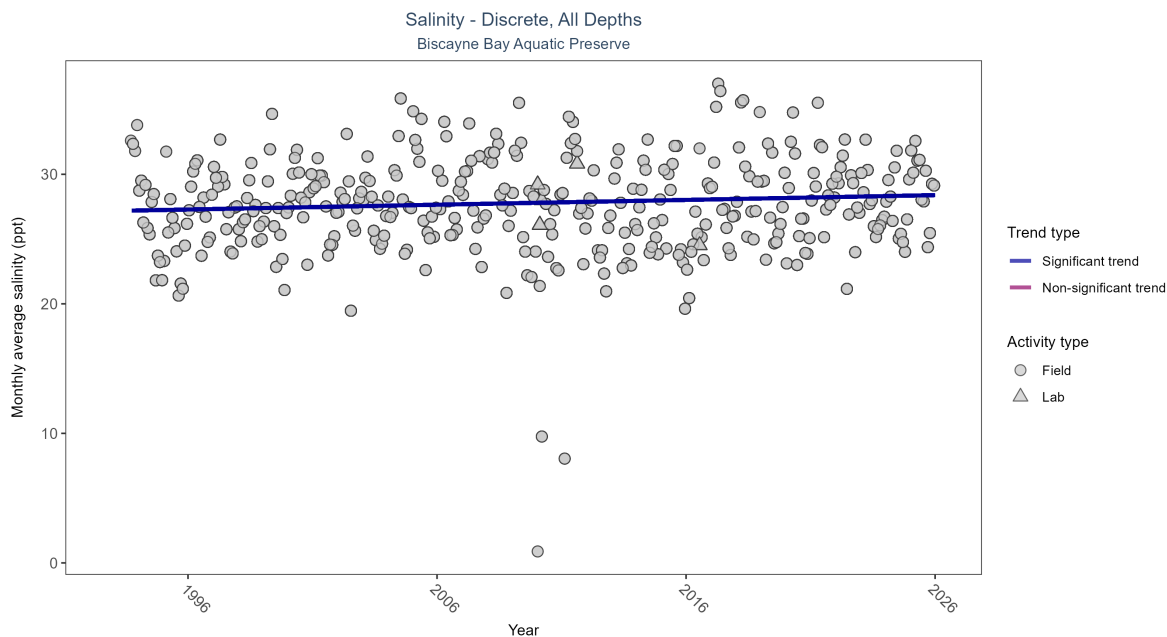


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly increasing trend	50529	33	1993 - 2025	31.4	0.089	27.16942	0.03703	0.013

Monthly average salinity increased by 0.04 ppt per year.

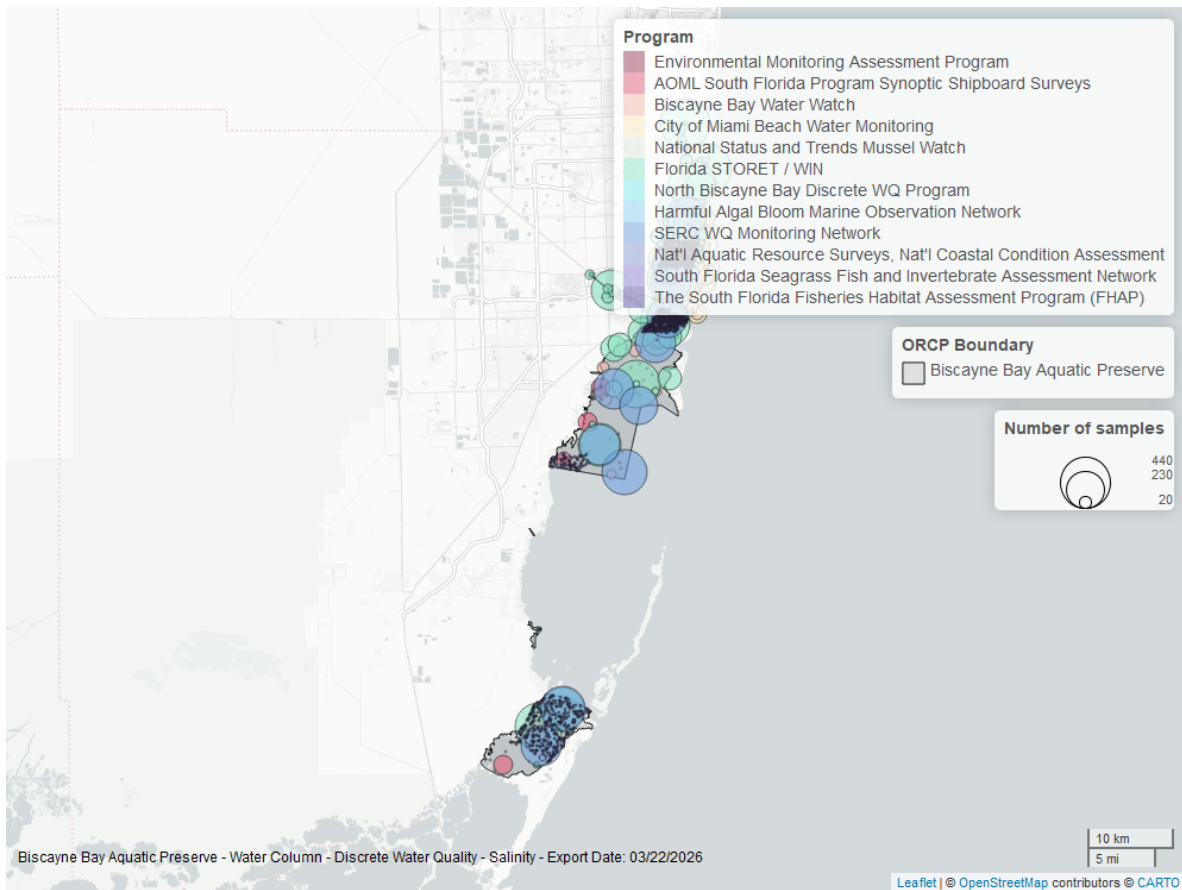


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

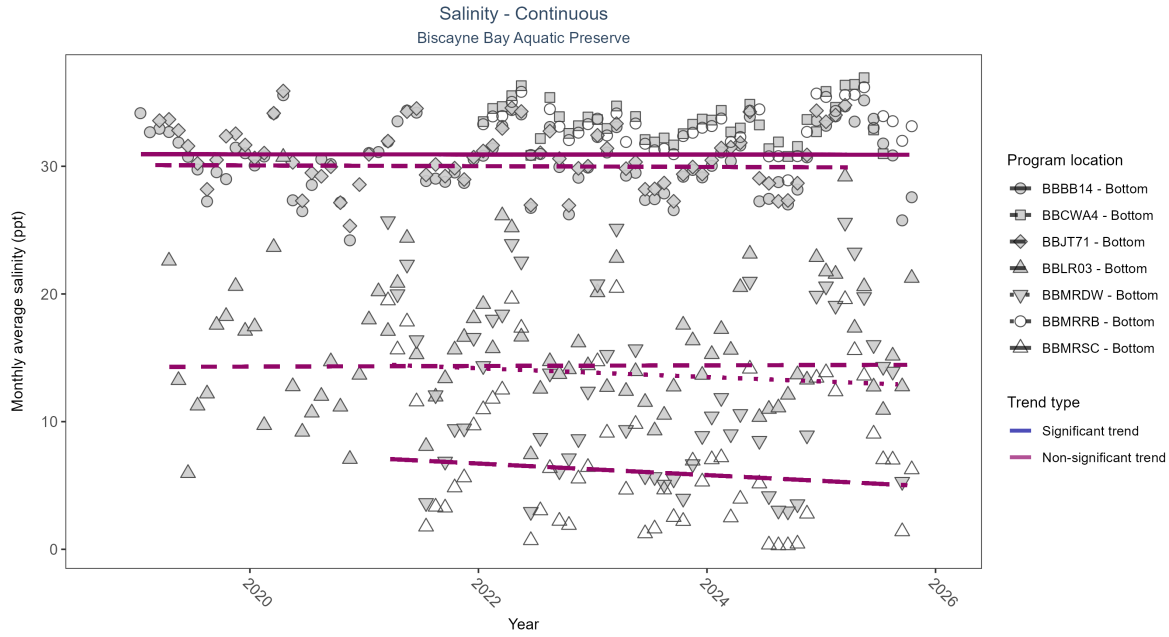


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	199380	7	2019 - 2025	30.6	-0.01	30.94	-0.01	1
BBCWA4	Insufficient data to calculate trend	106537	4	2022 - 2025	33.6	-	-	-	-
BBJT71	No significant trend	193275	7	2019 - 2025	31.0	-0.02	30.1	-0.03	0.8694
BBLR03	No significant trend	197528	7	2019 - 2025	18.0	0.02	14.29	0.02	0.9625
BBMRDW	No significant trend	146988	5	2021 - 2025	10.9	-0.17	14.52	-0.34	0.2357
BBMRRB	Insufficient data to calculate trend	124795	4	2022 - 2025	33.3	-	-	-	-
BBMRSC	No significant trend	153400	5	2021 - 2025	5.1	-0.13	7.17	-0.45	0.3159

No detectable change in monthly average salinity was observed at five locations. There was insufficient data to fit a model for two locations.

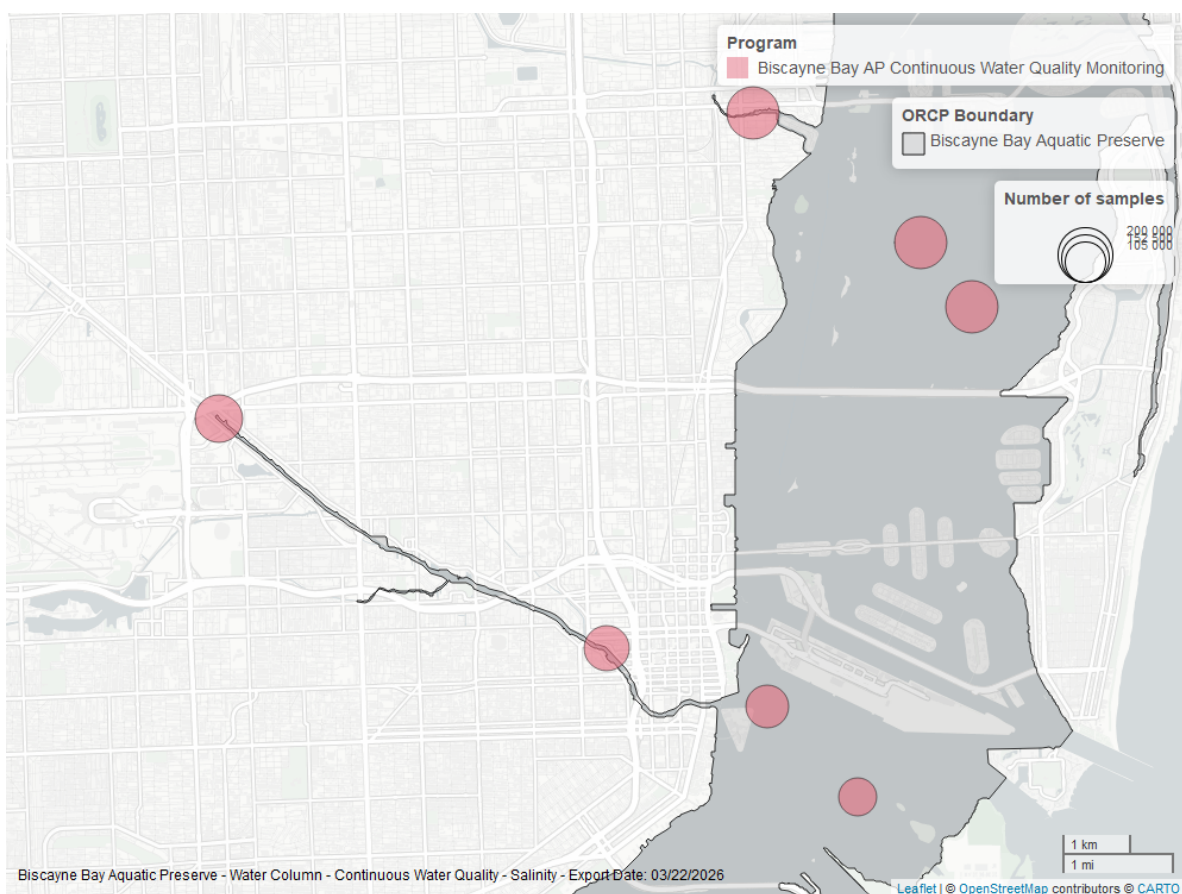


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

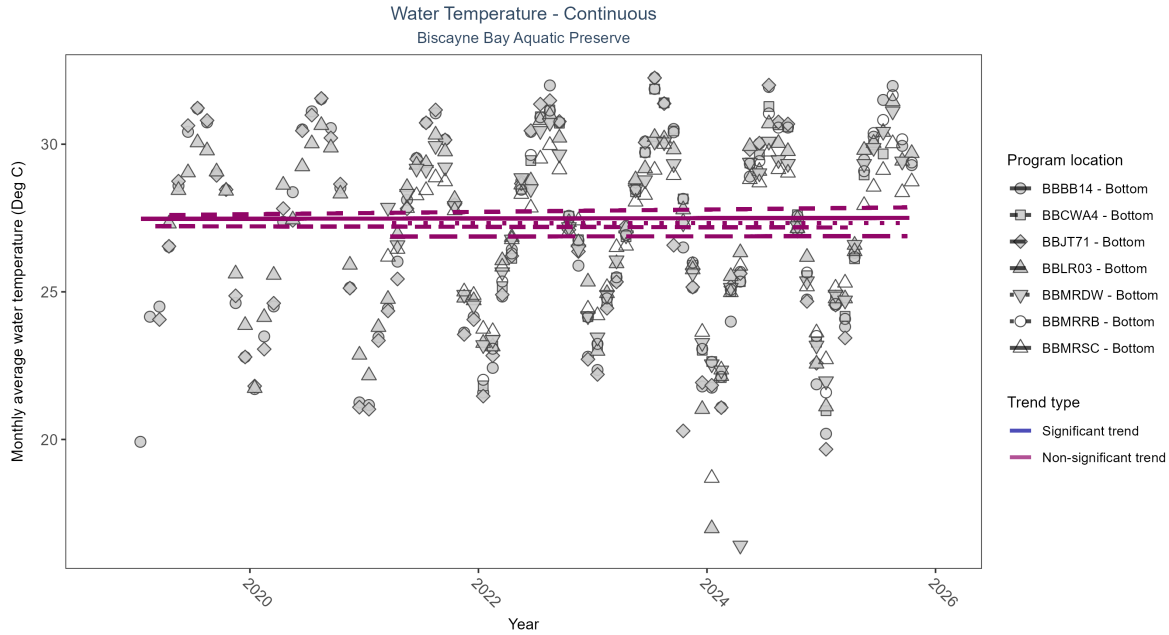


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	209878	7	2019 - 2025	27.3	0.02	27.47	0	0.8915
BBCWA4	Insufficient data to calculate trend	113090	4	2022 - 2025	27.1	-	-	-	-
BBJT71	No significant trend	208542	7	2019 - 2025	27.1	-0.03	27.22	-0.01	0.8737
BBLR03	No significant trend	206286	7	2019 - 2025	27.9	0.04	27.59	0.04	0.6721
BBMRDW	No significant trend	150209	5	2021 - 2025	27.6	-0.03	27.32	0	1
BBMRRB	Insufficient data to calculate trend	125603	4	2022 - 2025	27.3	-	-	-	-
BBMRSC	No significant trend	154183	5	2021 - 2025	27.3	-0.02	26.87	0	1

No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for two locations.

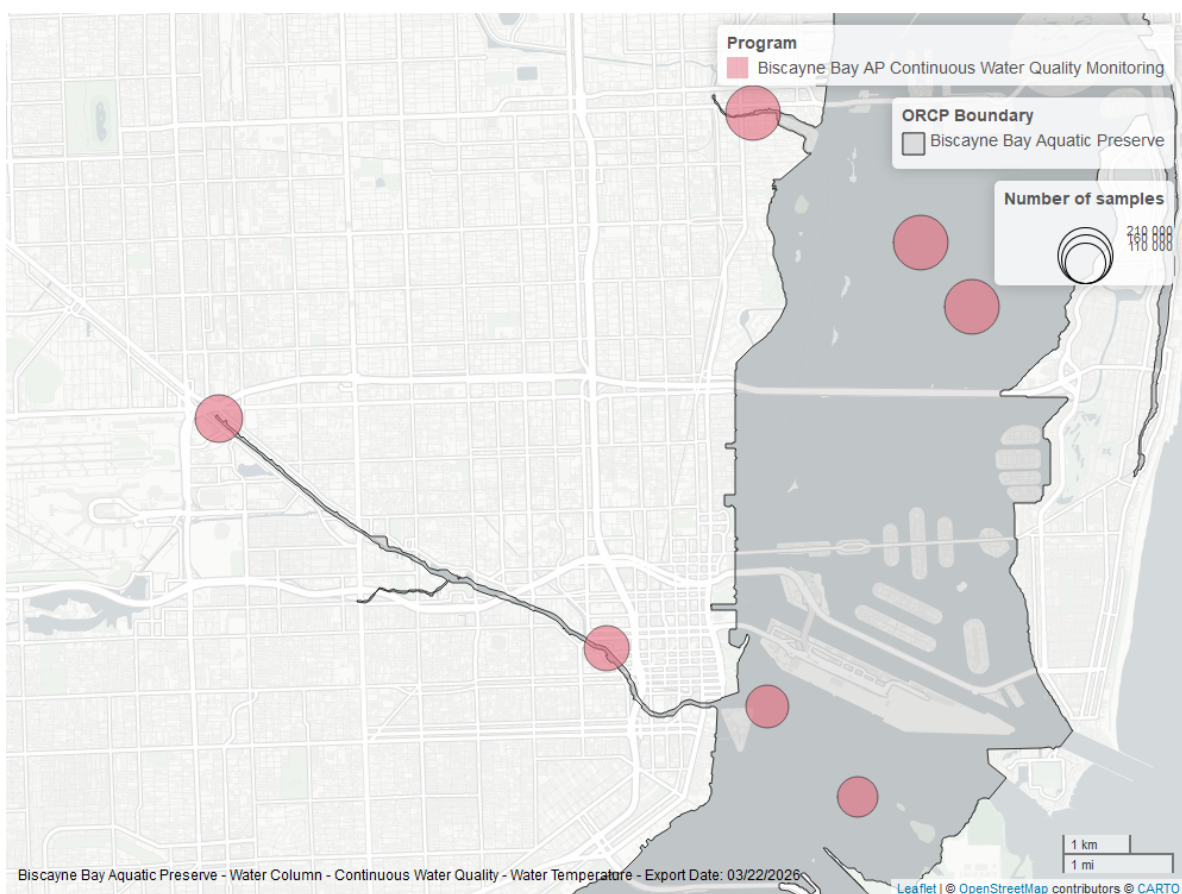


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

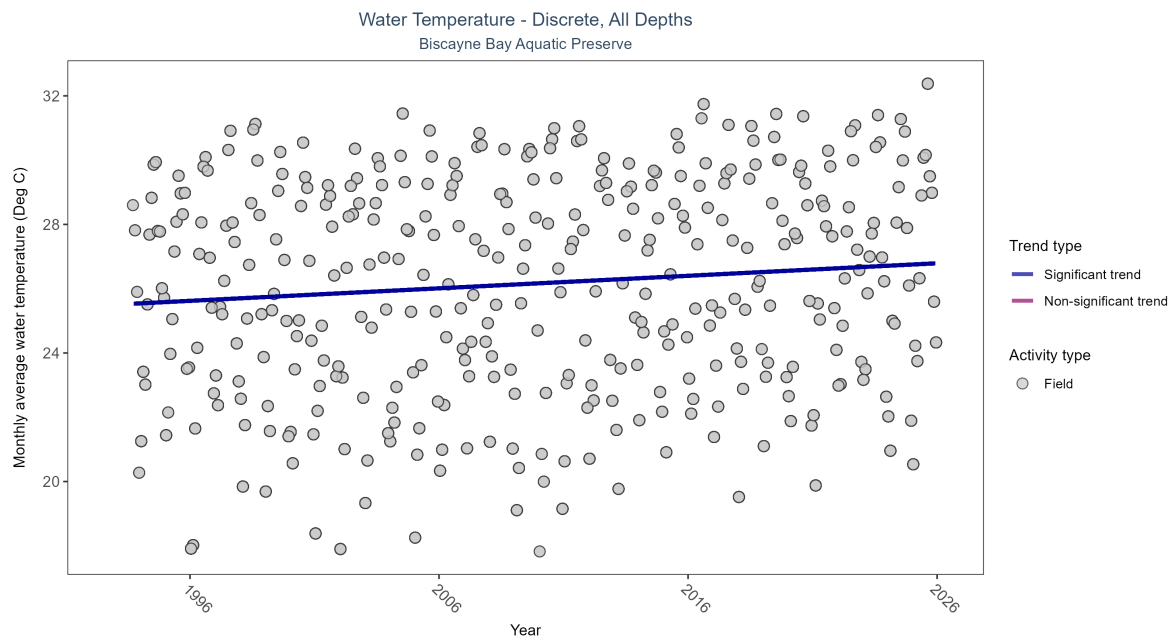


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	50728	33	1993 - 2025	26.63	0.22766	25.50481	0.03897	0

Monthly average water temperature increased by 0.04°C per year.

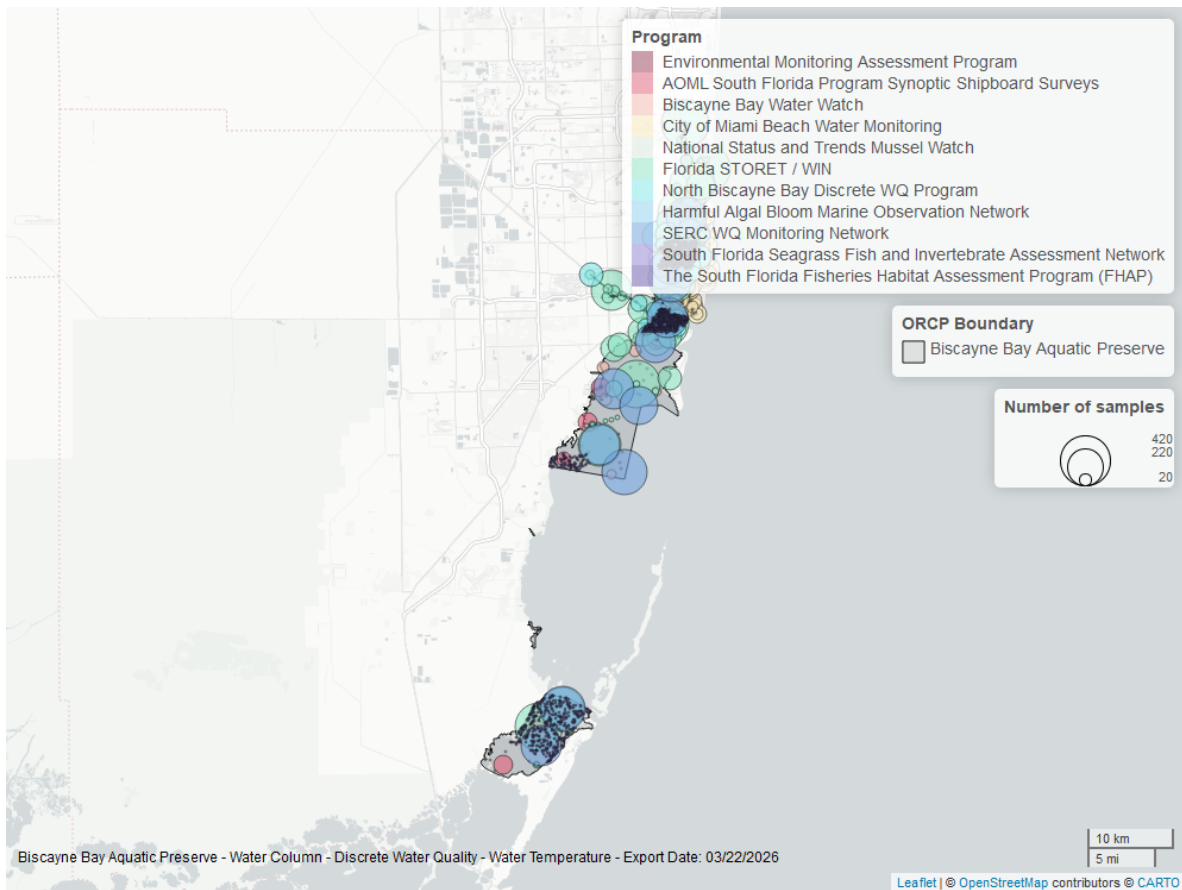


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

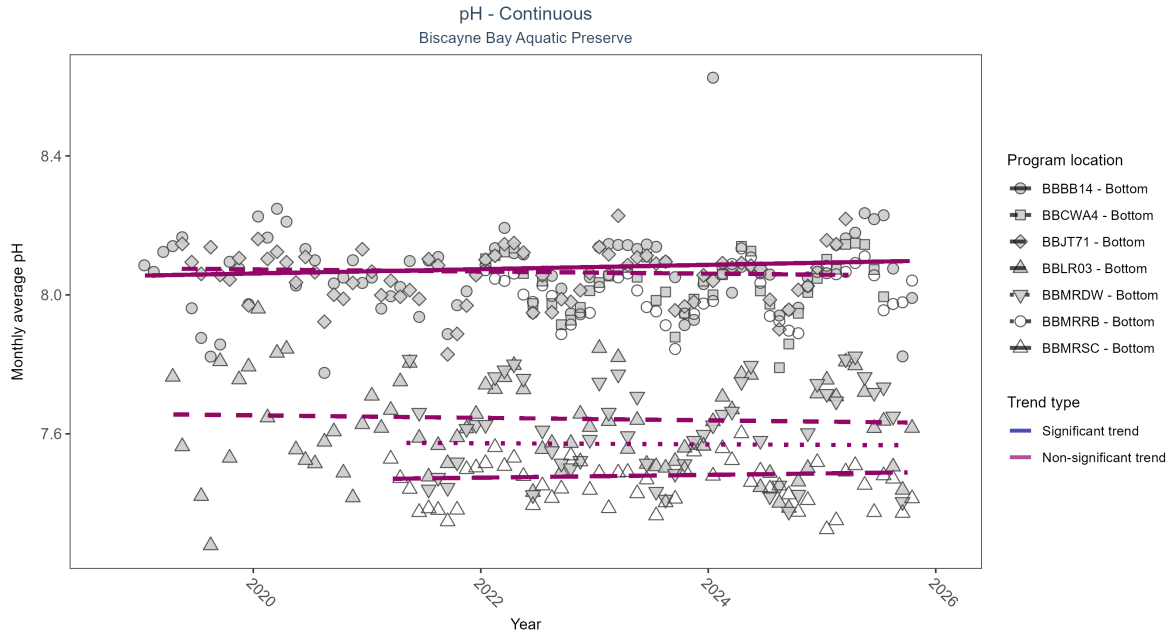


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	205693	7	2019 - 2025	8.1	0.1	8.06	0.01	0.3398
BBCWA4	Insufficient data to calculate trend	112102	4	2022 - 2025	8.0	-	-	-	-
BBJT71	No significant trend	195449	7	2019 - 2025	8.1	-0.07	8.08	0	0.5078
BBLR03	No significant trend	195530	7	2019 - 2025	7.6	-0.05	7.66	0	0.6665
BBMRDW	No significant trend	145249	5	2021 - 2025	7.6	-0.03	7.57	0	0.9336
BBMRRB	Insufficient data to calculate trend	124839	4	2022 - 2025	8.0	-	-	-	-
BBMRSC	No significant trend	150527	5	2021 - 2025	7.4	0.03	7.47	0	0.6997

No detectable change in monthly average pH was observed at five locations. There was insufficient data to fit a model for two locations.

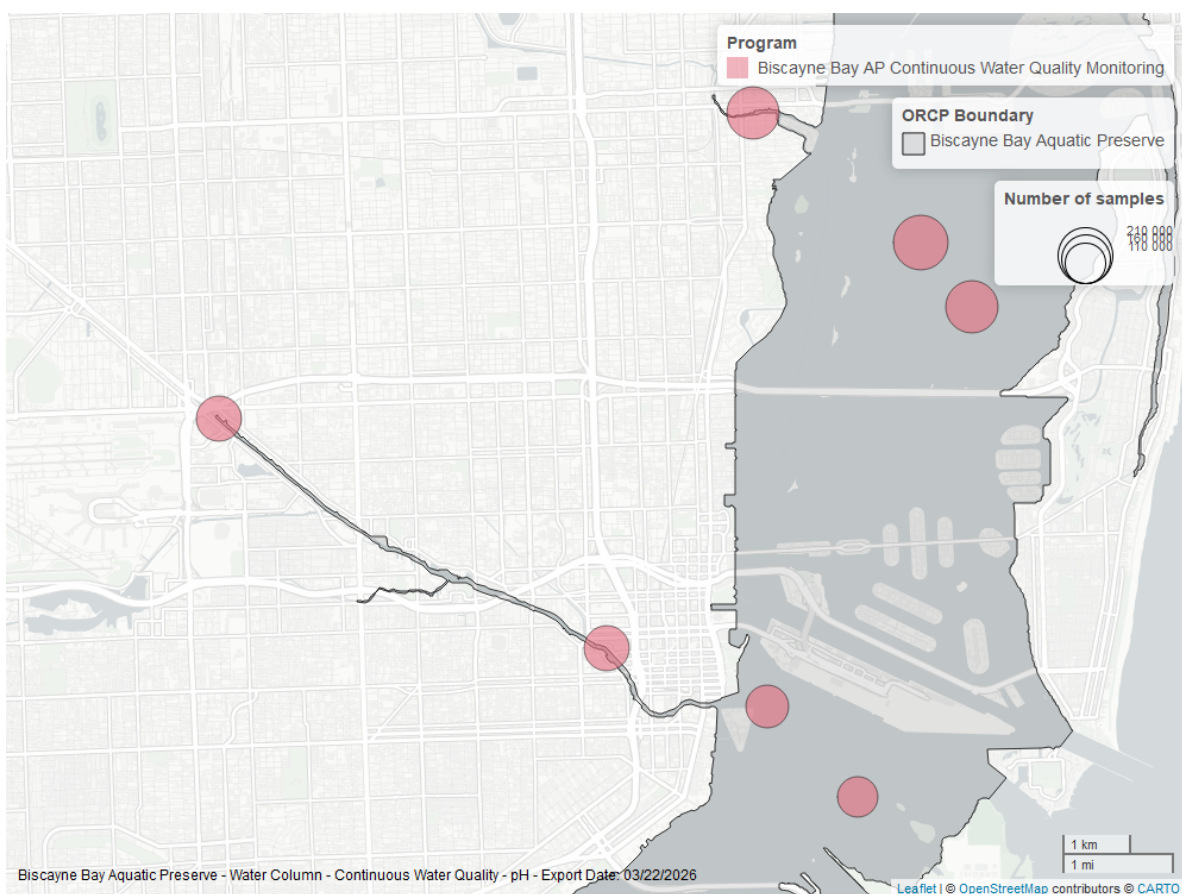


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

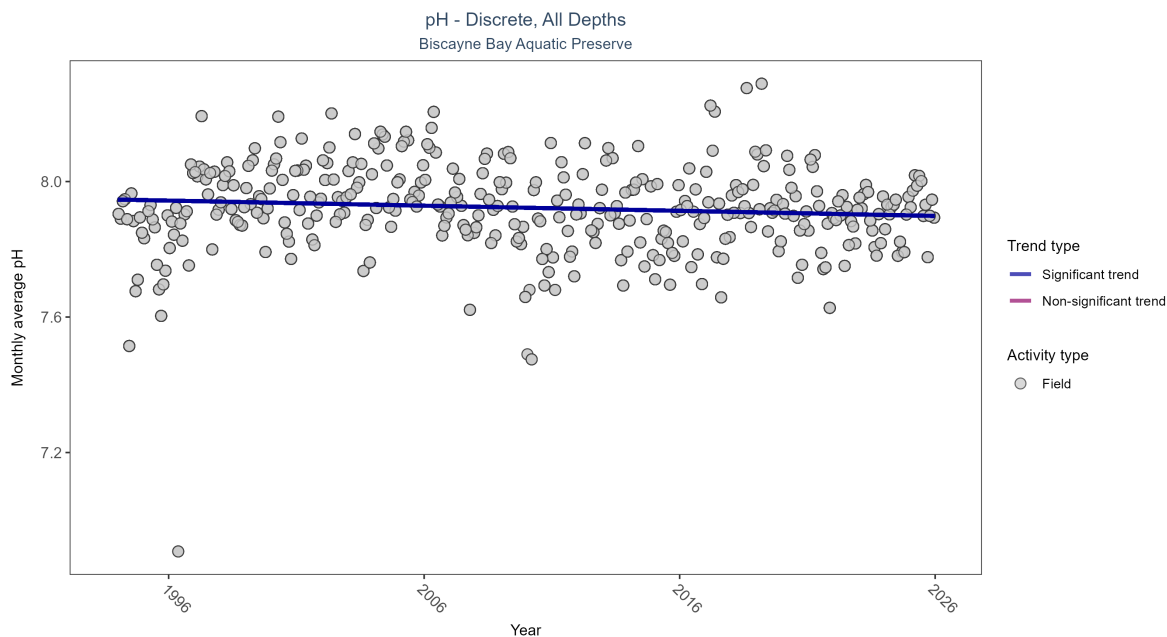


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	43873	32	1994 - 2025	7.98	-0.10437	7.94675	-0.00151	0.0038

Monthly average pH decreased by less than 0.01 pH units per year.

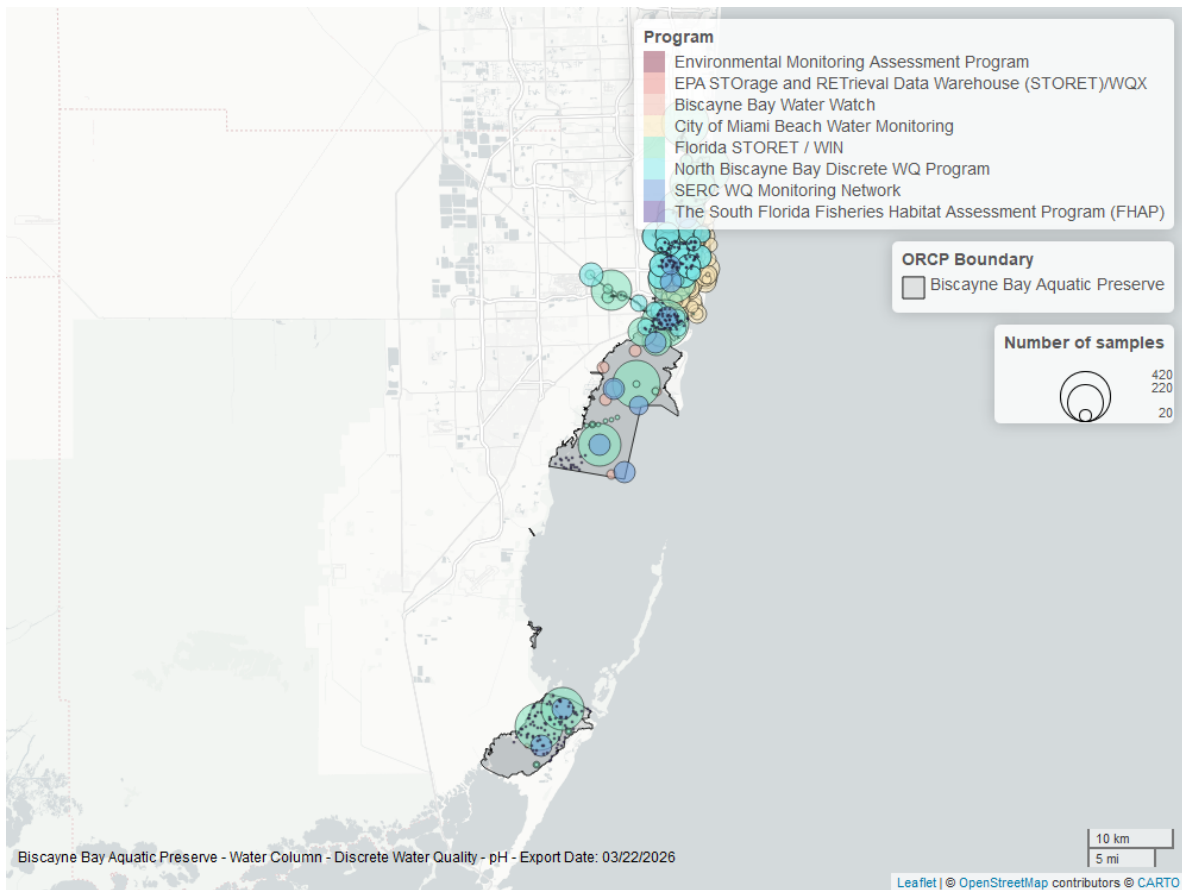


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

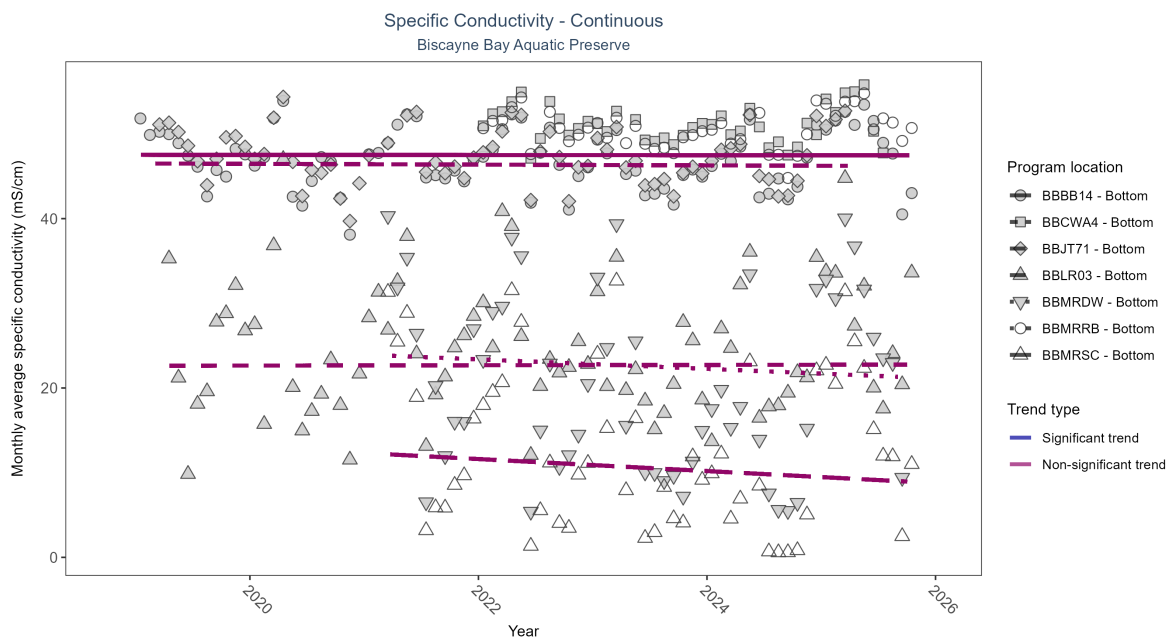


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	199381	7	2019 - 2025	47.11	-0.02	47.54	-0.01	0.854
BBCWA4	Insufficient data to calculate trend	106556	4	2022 - 2025	51.22	-	-	-	-
BBJT71	No significant trend	193275	7	2019 - 2025	47.63	-0.01	46.51	-0.05	0.9563
BBLR03	No significant trend	199348	7	2019 - 2025	28.44	0.01	22.63	0.02	1
BBMRDW	No significant trend	146990	5	2021 - 2025	18.45	-0.17	23.94	-0.56	0.2357
BBMRRB	Insufficient data to calculate trend	124795	4	2022 - 2025	50.80	-	-	-	-
BBMRSC	No significant trend	154166	5	2021 - 2025	8.98	-0.09	12.32	-0.71	0.4875

No detectable change in monthly average specific conductivity was observed at five locations. There was insufficient data to fit a model for two locations.

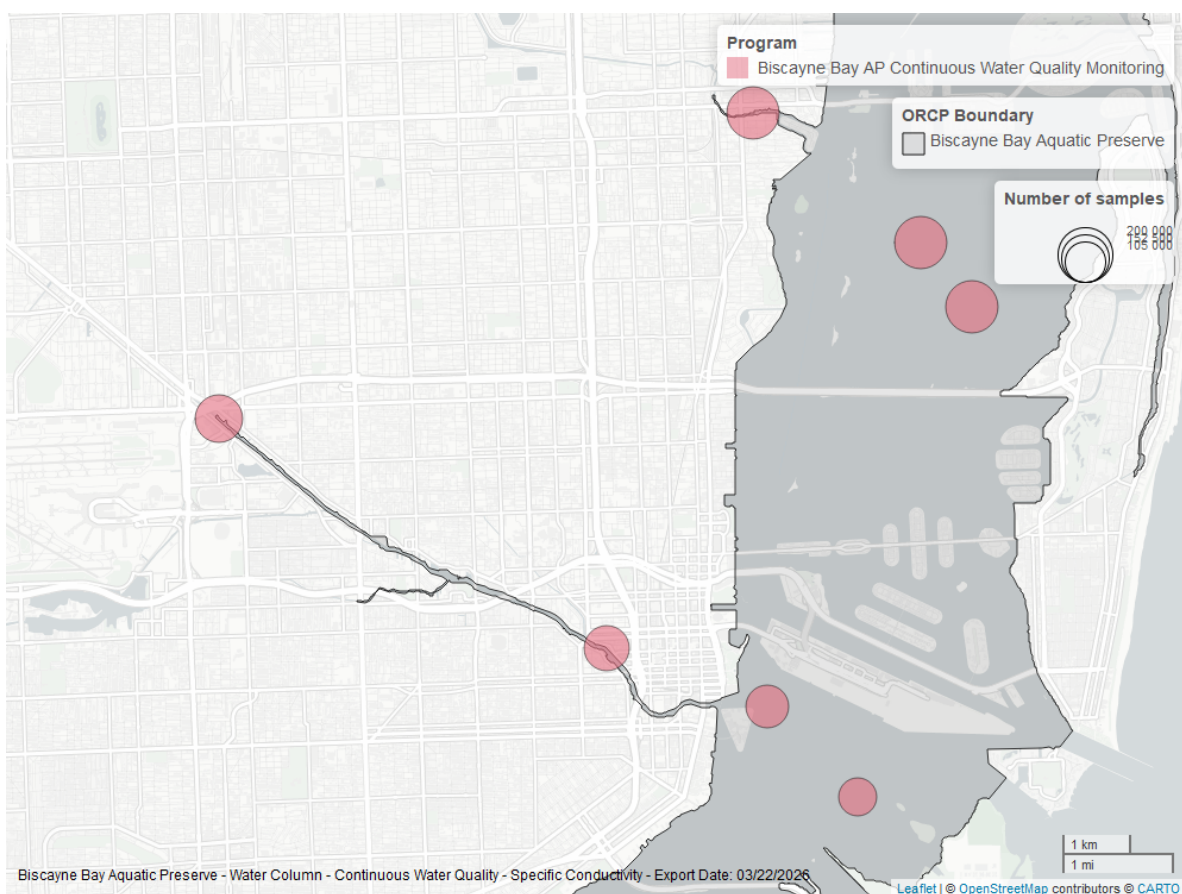


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

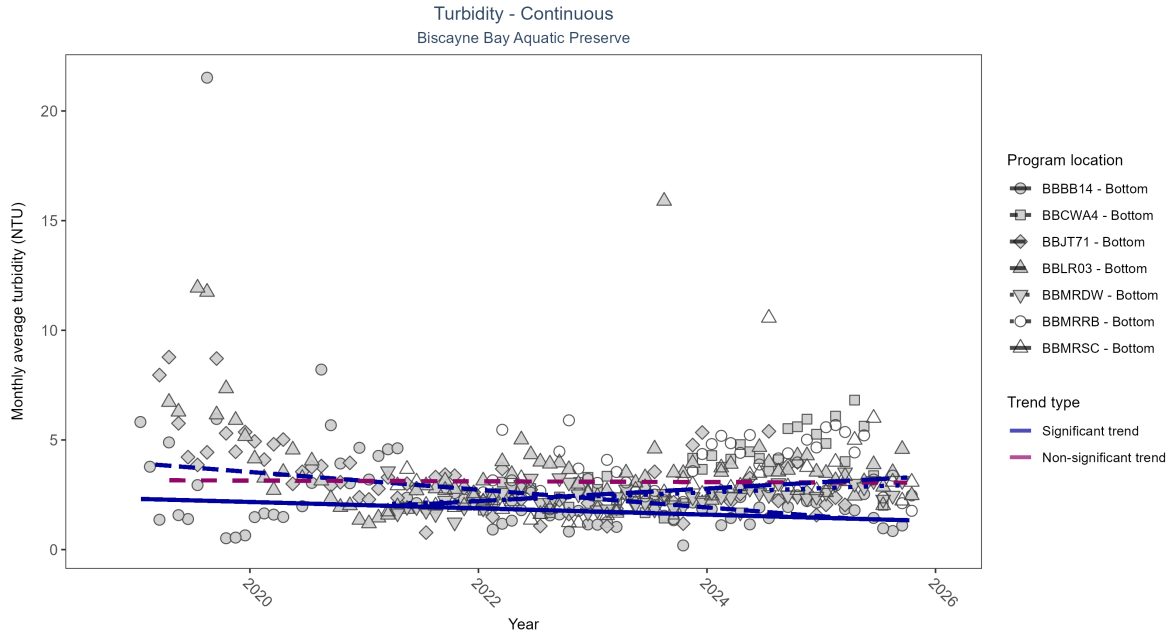


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	Significantly decreasing trend	195719	7	2019 - 2025	2	-0.22	2.32	-0.14	0.0184
BBCWA4	Insufficient data to calculate trend	109259	4	2022 - 2025	3	-	-	-	-
BBJT71	Significantly decreasing trend	201128	7	2019 - 2025	3	-0.46	3.94	-0.4	0
BBLR03	No significant trend	202449	7	2019 - 2025	3	-0.01	3.16	-0.02	0.8108
BBMRDW	Significantly increasing trend	147146	5	2021 - 2025	2	0.35	2.04	0.19	0.0091
BBMRRB	Insufficient data to calculate trend	123415	4	2022 - 2025	3	-	-	-	-
BBMRSC	Significantly increasing trend	140544	5	2021 - 2025	2	0.41	1.92	0.29	0.002

At two program locations, monthly average turbidity increased by 0.19 NTU per year at one site and by 0.29 NTU per year at the other. At two program locations, monthly average turbidity decreased by 0.14 NTU per year at one site and by 0.4 NTU per year at the other. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for two locations.

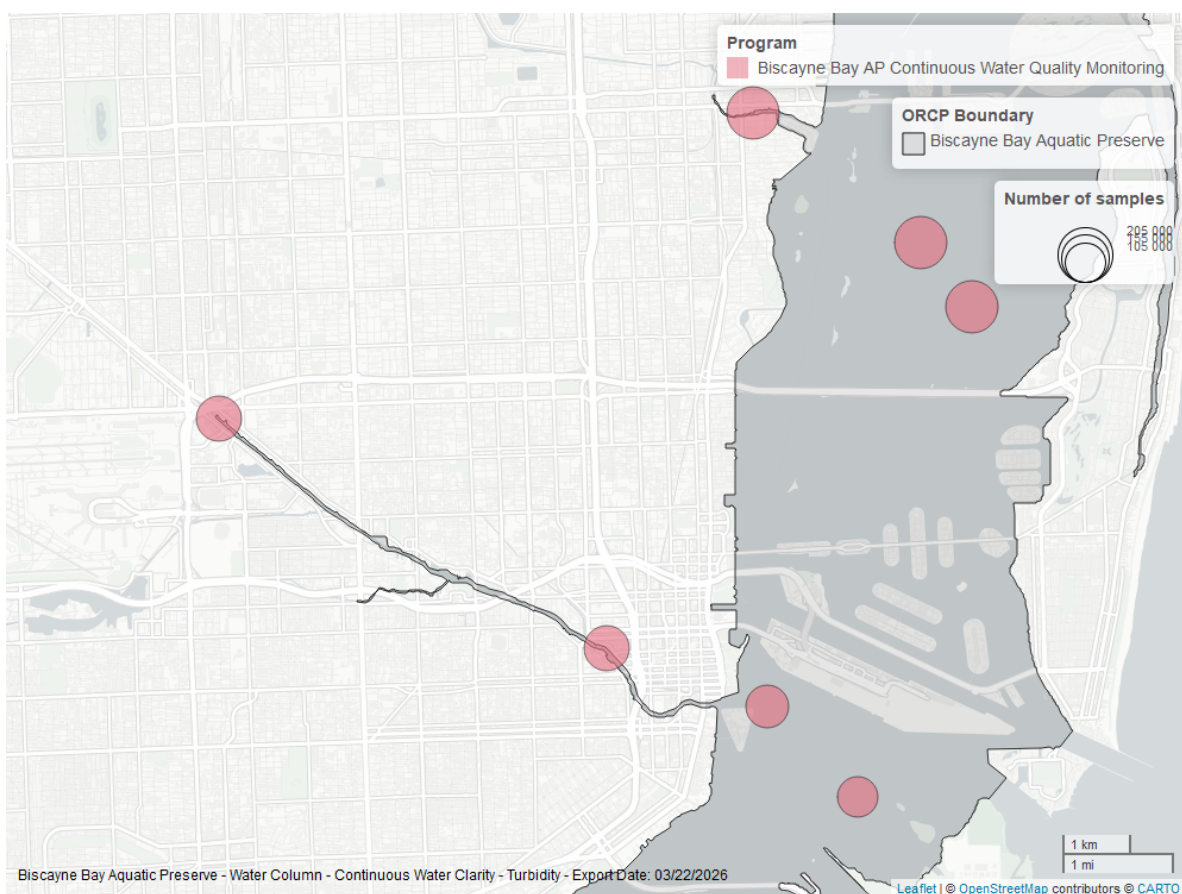


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

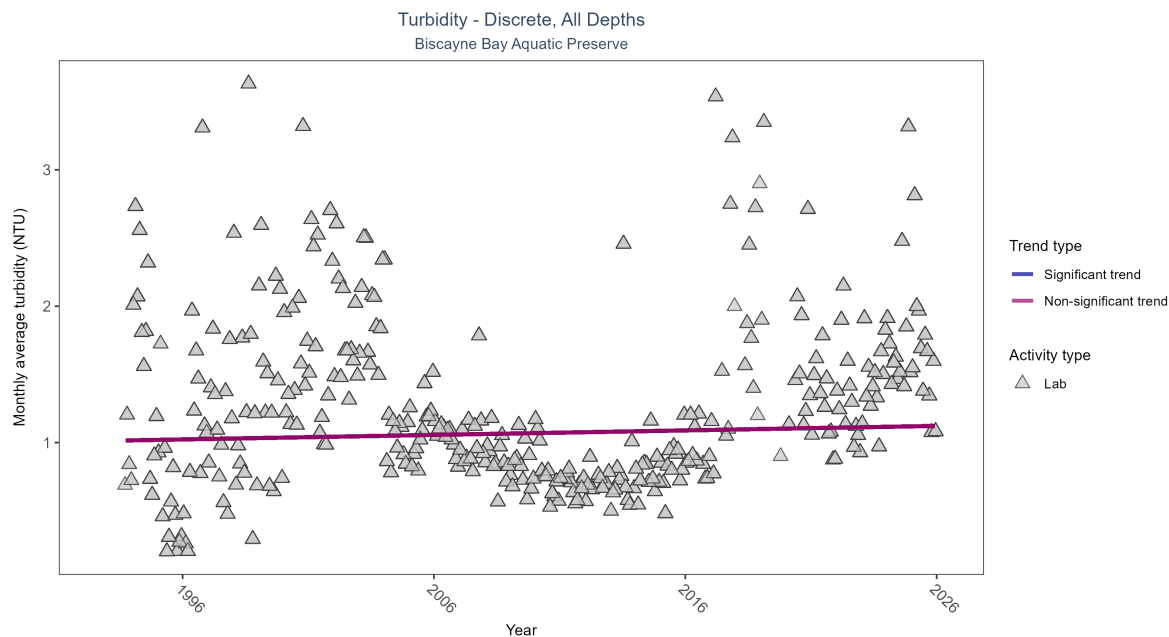


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	13958	33	1993 - 2025	0.9	0.03996	1.01324	0.00331	0.2562

Turbidity showed no detectable trend between 1993 and 2025.

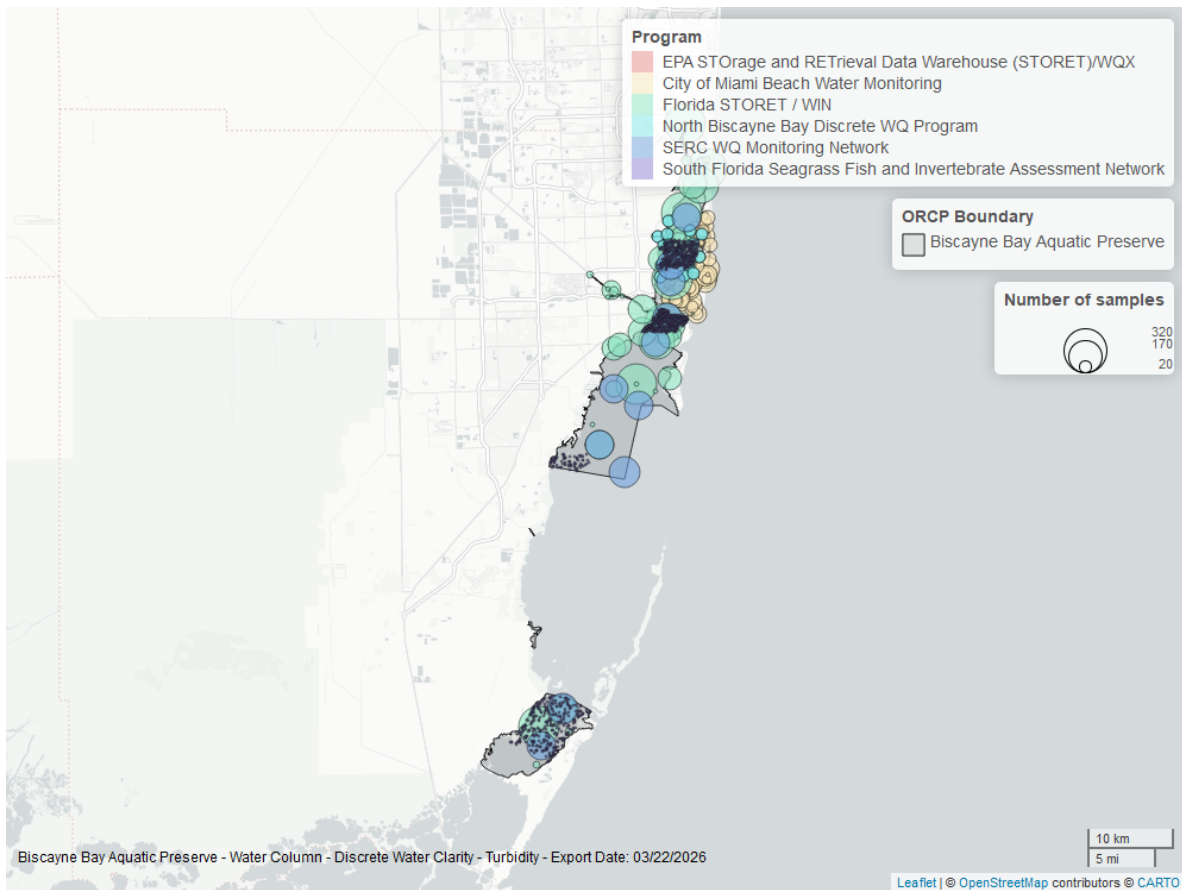


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

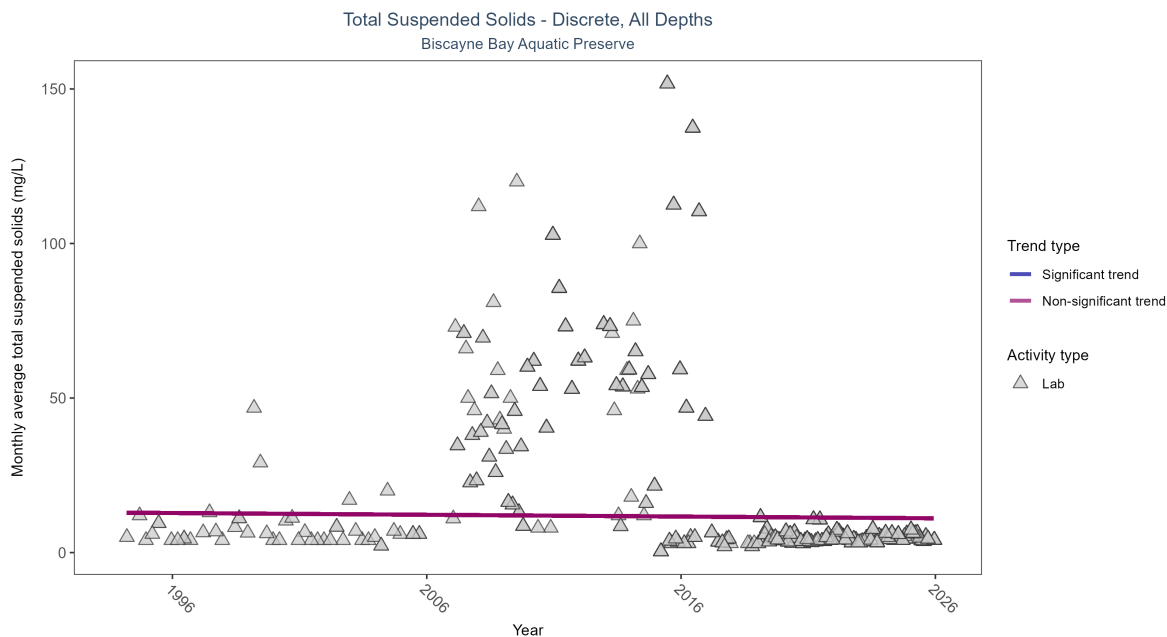


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	2900	31	1994 - 2025	5	-0.14792	12.91189	-0.05705	0.1201

Total suspended solids showed no detectable trend between 1994 and 2025.

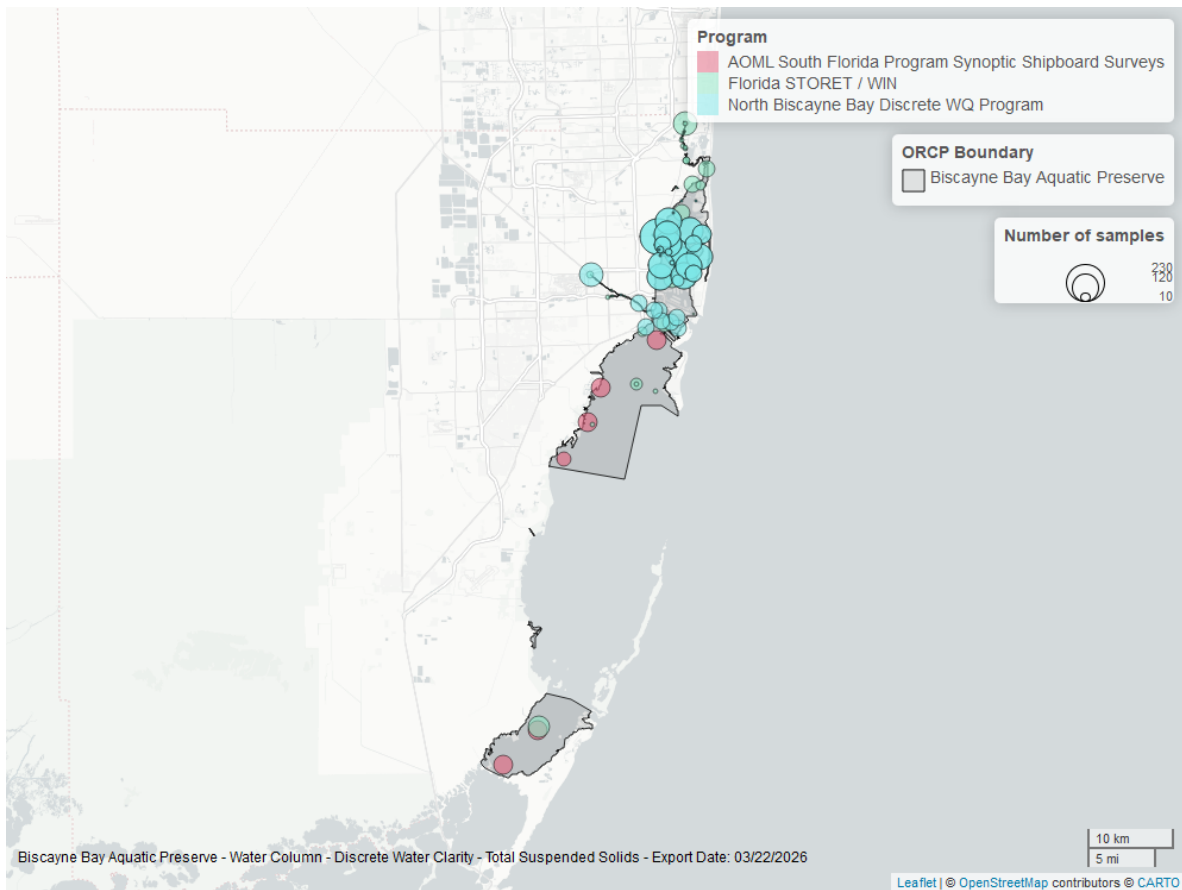


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

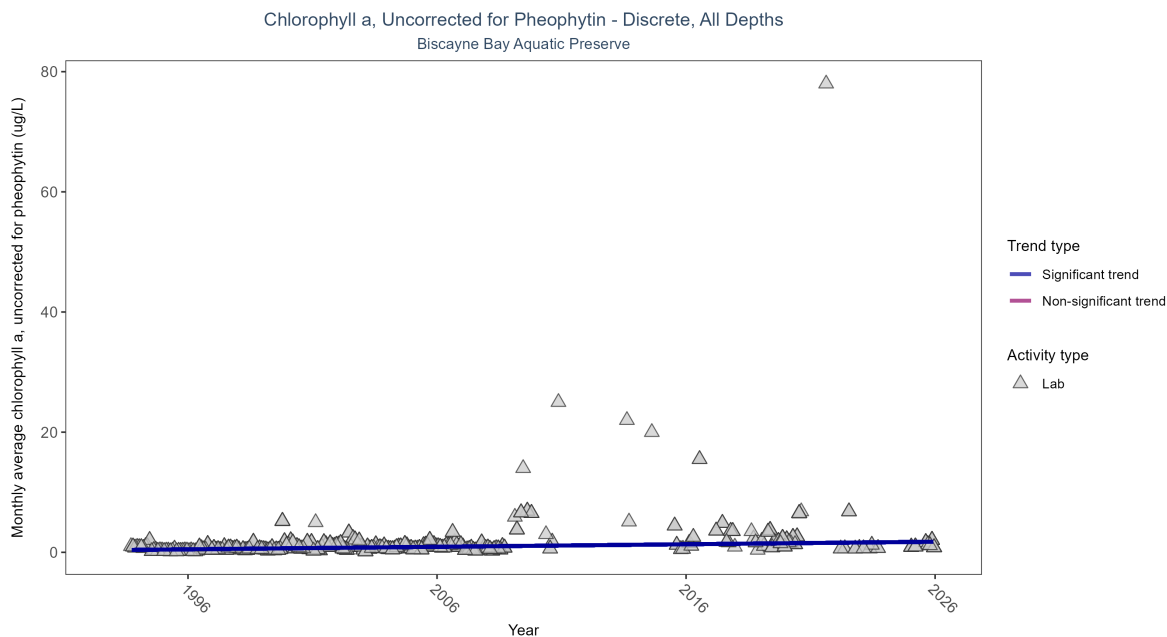


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	2517	30	1993 - 2025	0.70231	0.37392	0.37401	0.04208	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.04 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

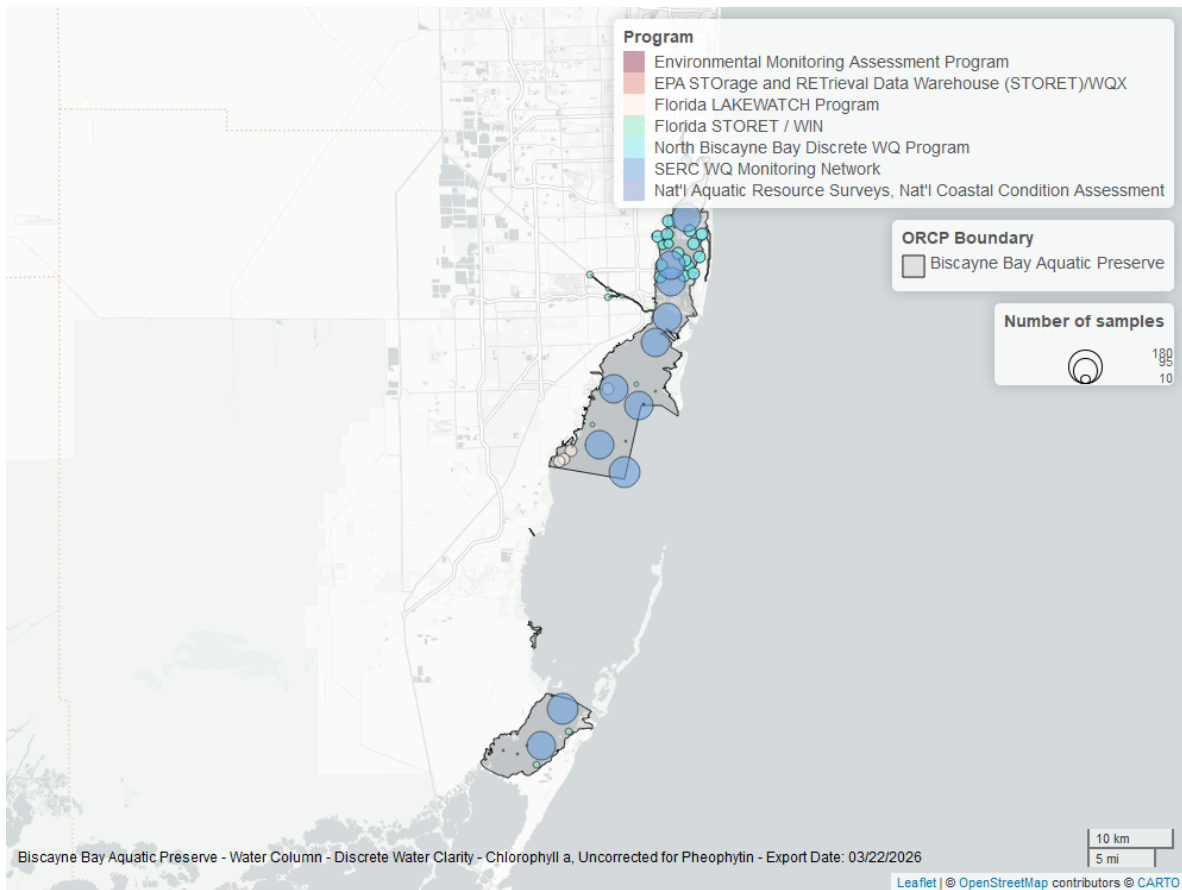


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

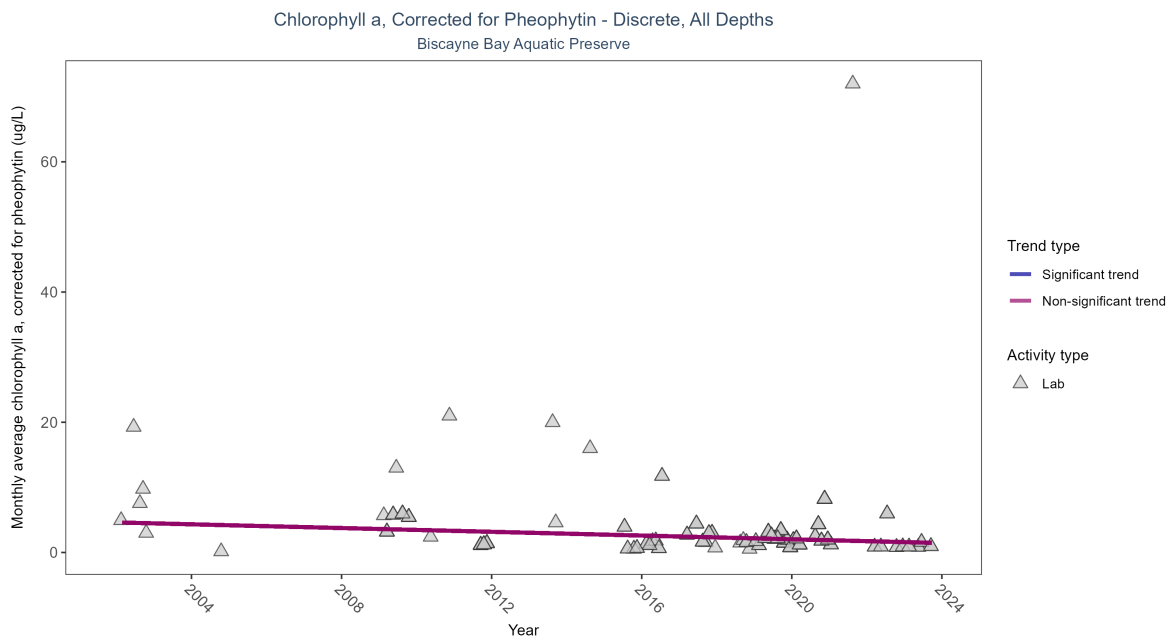


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	492	16	2002 - 2023	1.4	-0.20161	4.61564	-0.14393	0.0505

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2023.

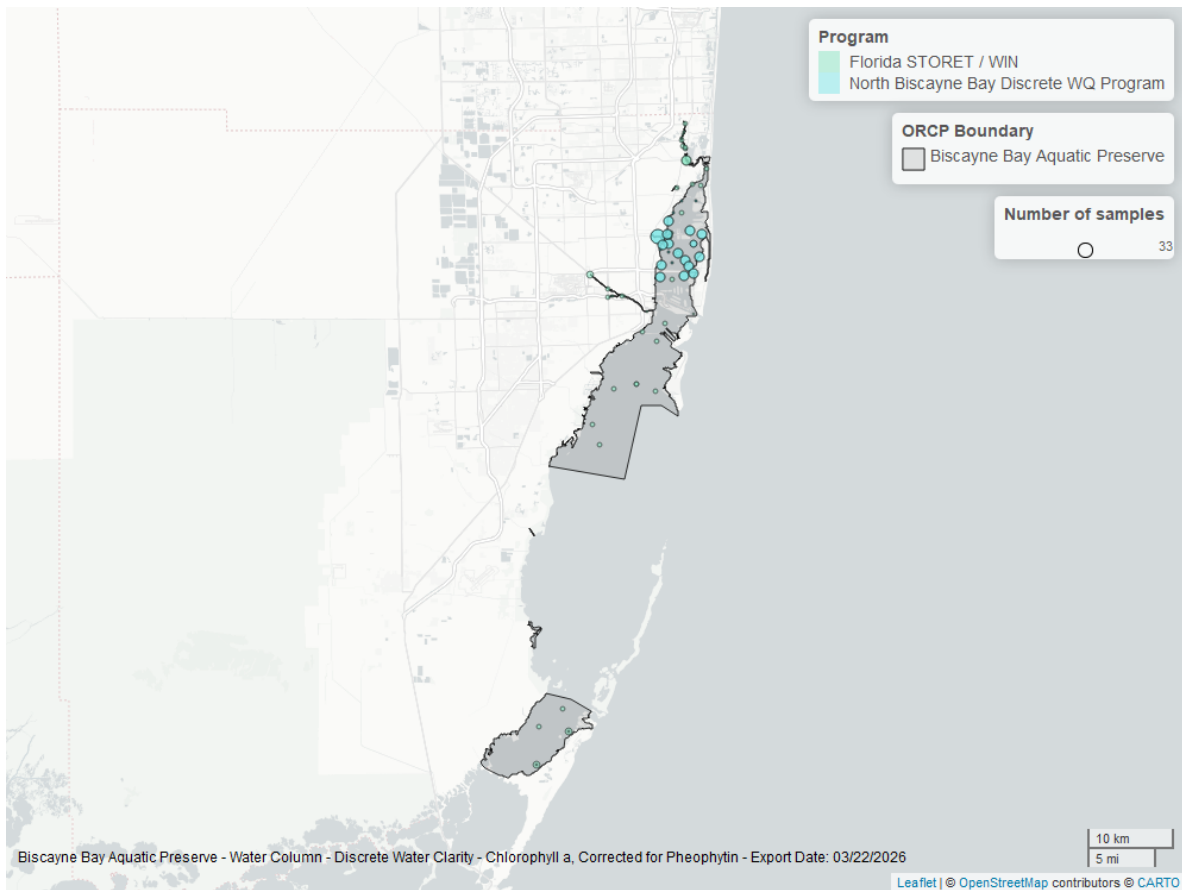


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

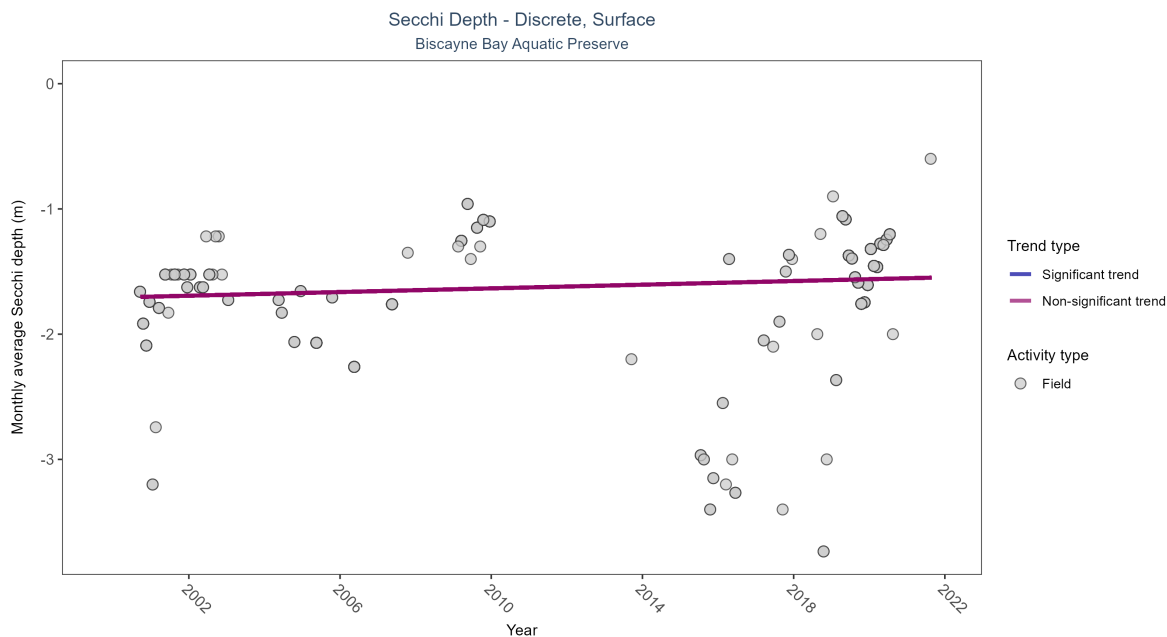


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	550	17	2000 - 2021	-1.45	0.14531	-1.70768	0.00733	0.4217

Secchi depth showed no detectable trend between 2000 and 2022.

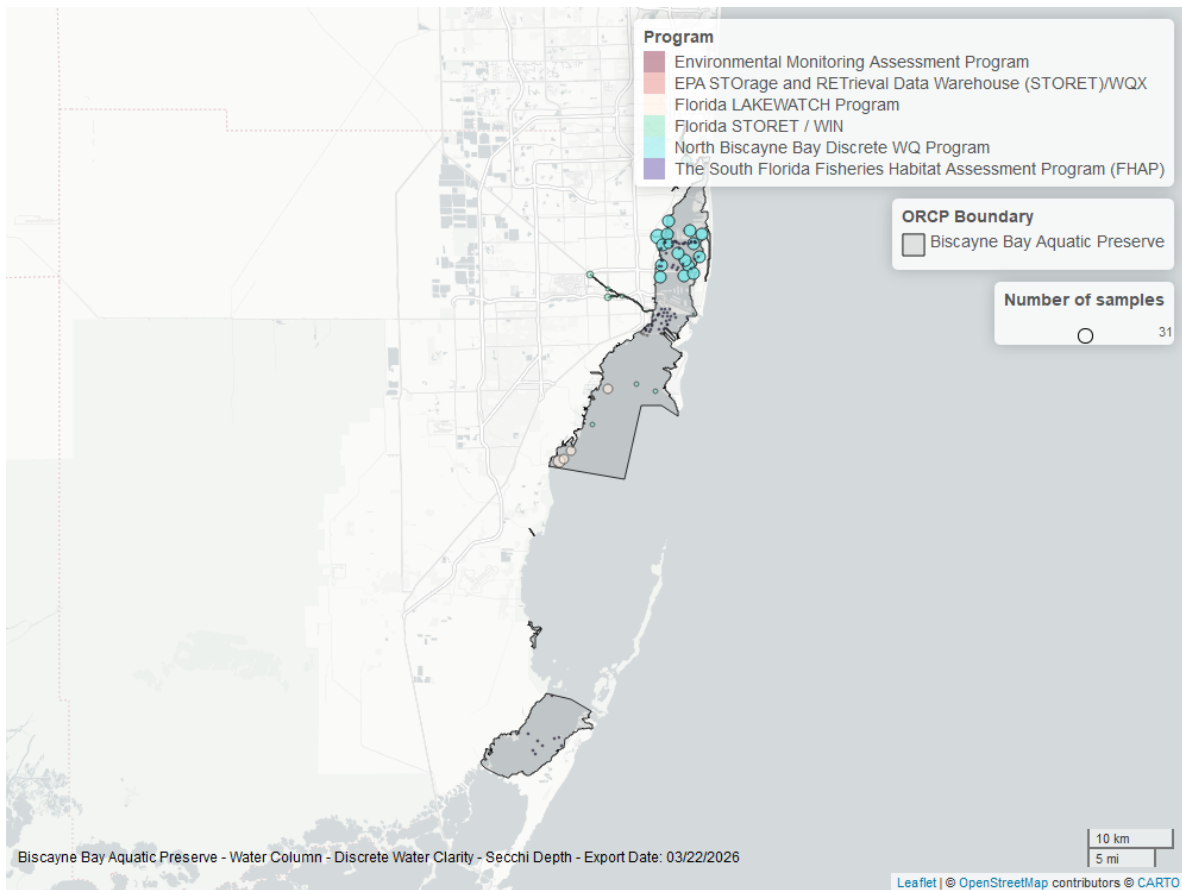


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

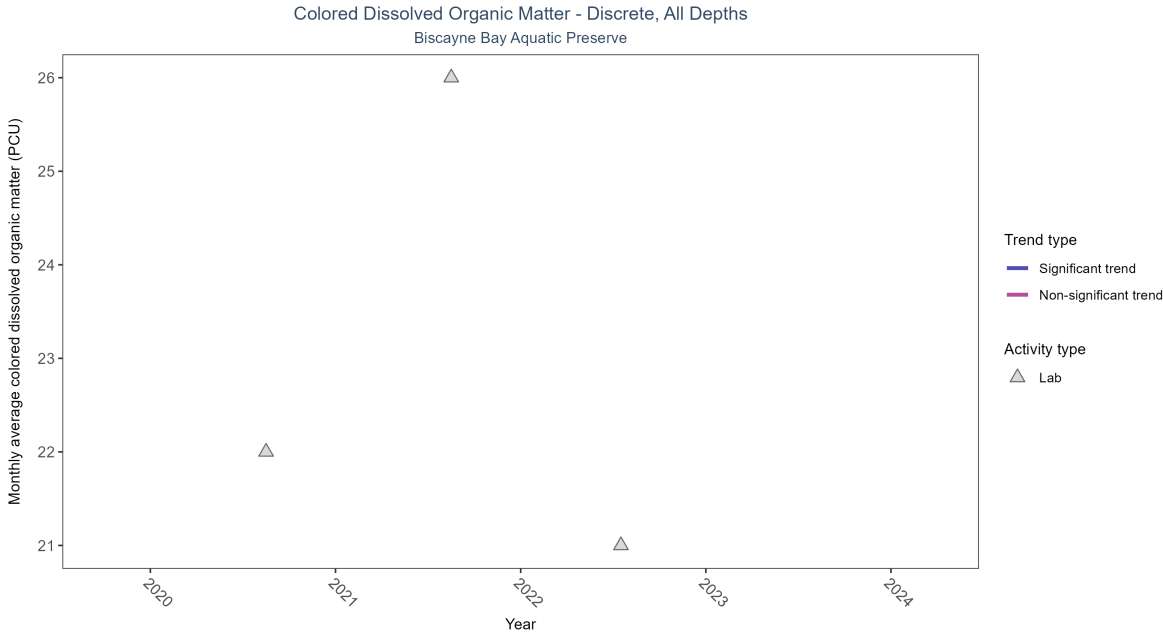


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	3	3	2020 - 2022	22	-	-	-	-

There was insufficient data to fit a model for colored dissolved organic matter.

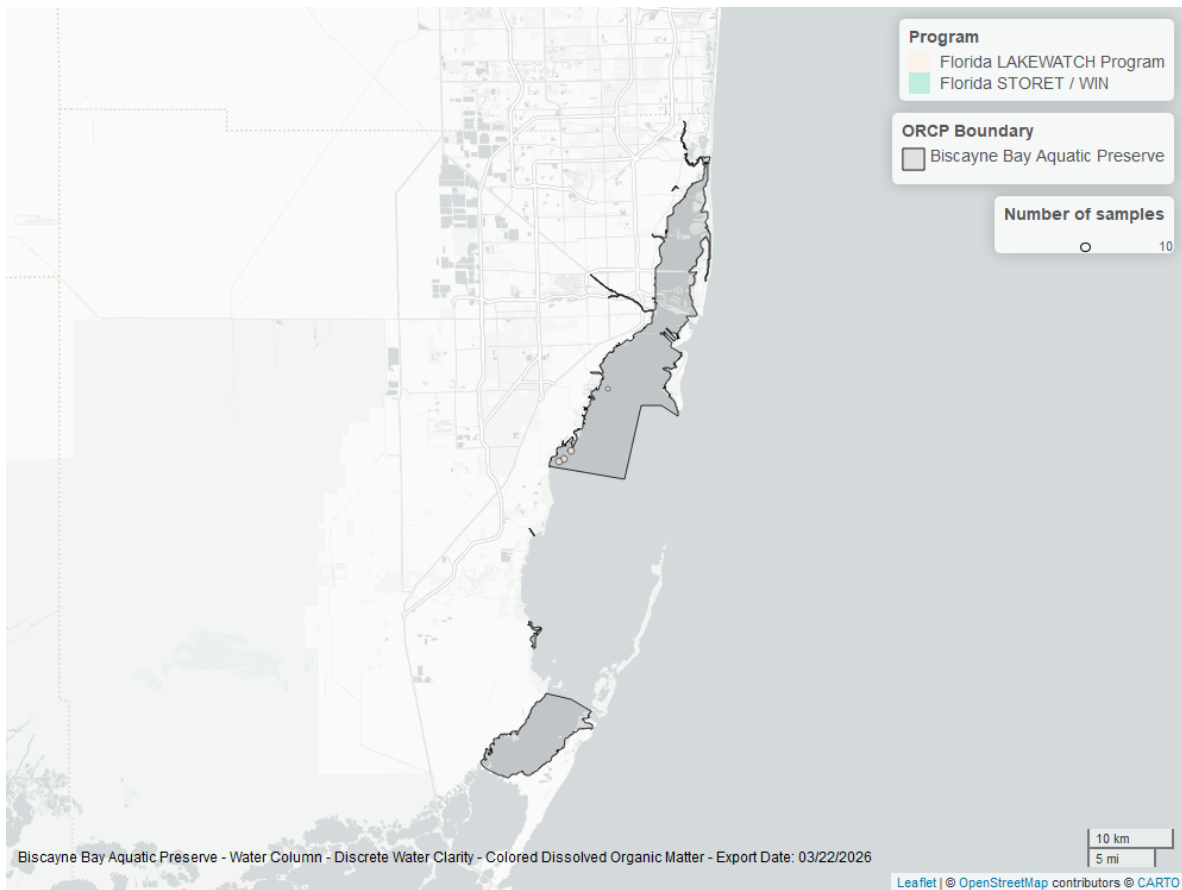


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.